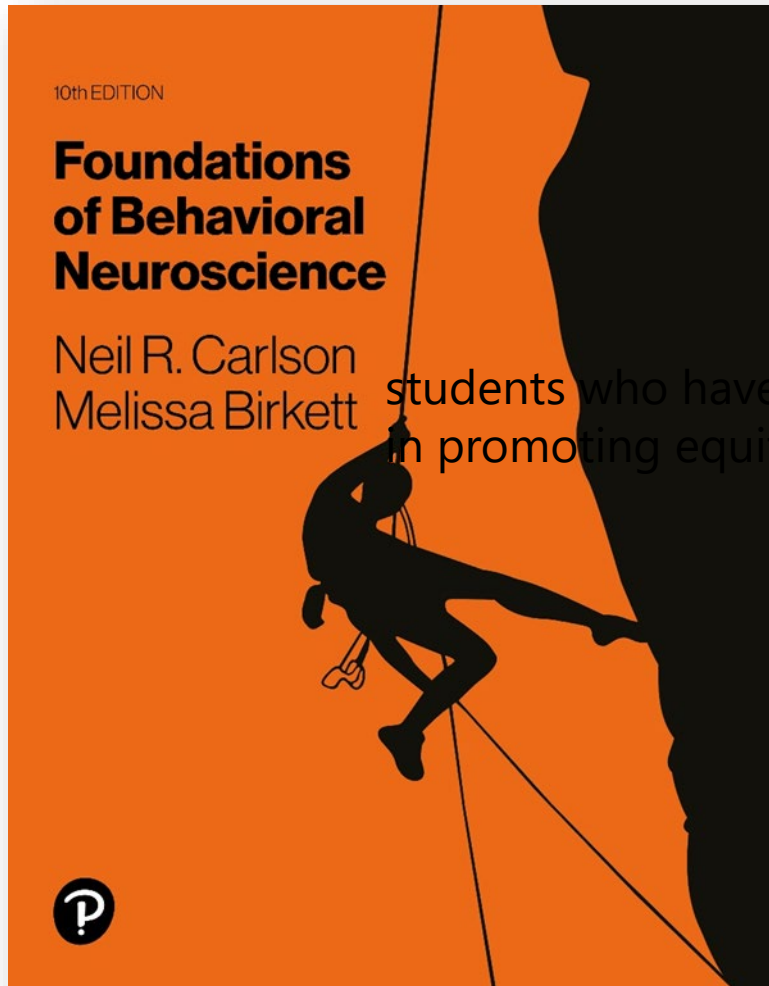


Foundations of Behavioral Neuroscience

Tenth Edition



Chapter 10

Sexual and Reproductive Behaviors

students who have demonstrated interest in promoting equity in the field of epilepsy.

< [Back to Course Offerings](#)

FRONTIERS IN REPRODUCTION

Overview

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Frontiers in Reproduction: Molecular and Cellular Concepts and Applications

Frontiers in Reproduction (FIR) is an intensive six-week laboratory and lecture course designed for advanced graduate students, post-doctoral fellows, newly independent scientists and physicians who seek training in modern state-of-the-art methods and a broad view of current concepts in all areas of reproductive biology.

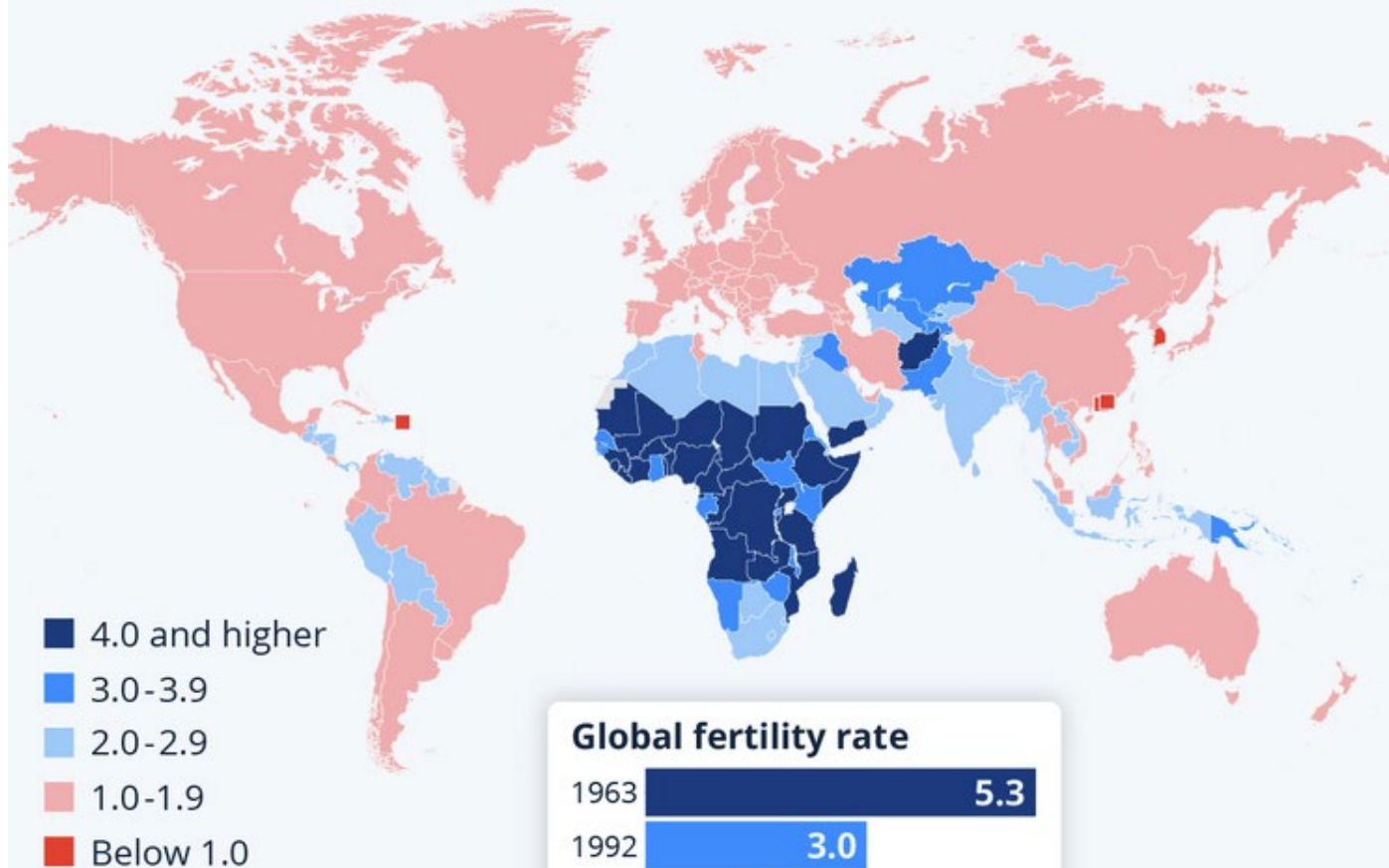
[Development and Reproduction](#)

Course/Program Dates: Apr 26, 2026 - Jun 07, 2026

Application due date: Jan 12, 2026

The State of Global Fertility

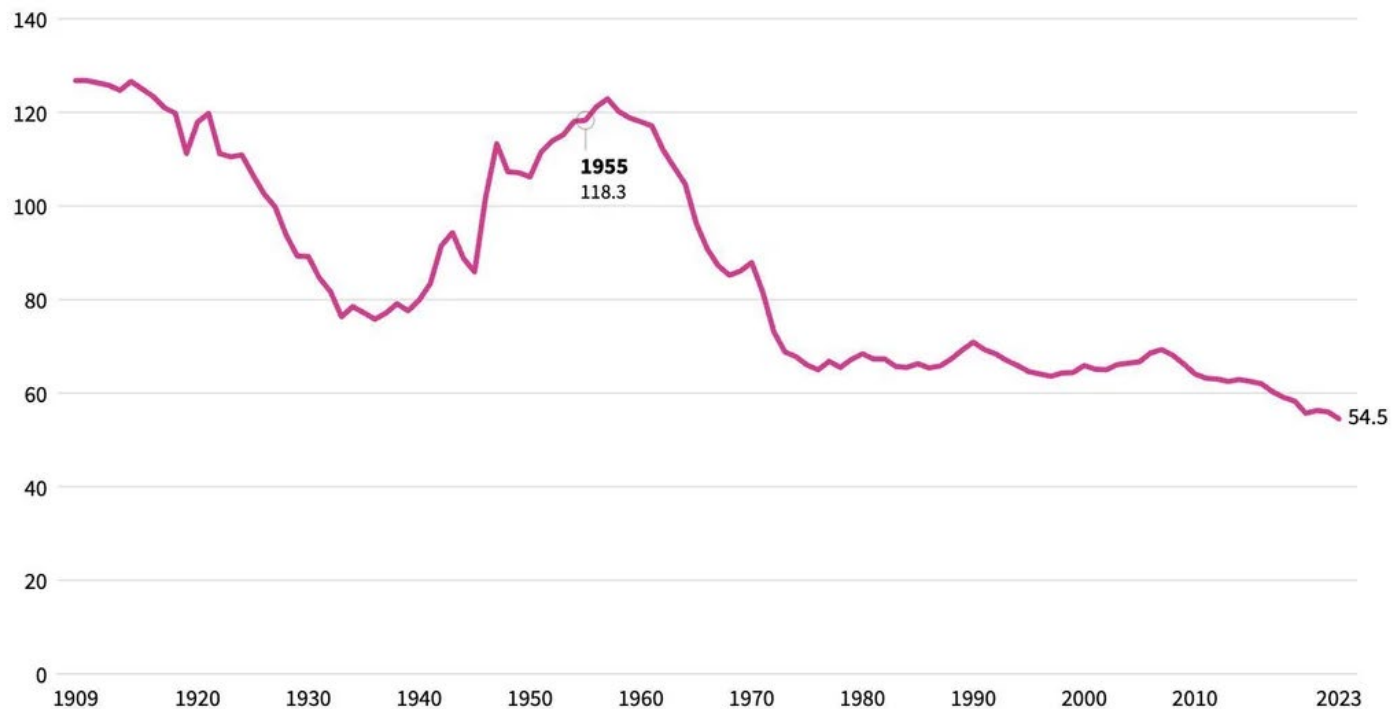
Fertility rate by country in 2023 (births per woman)



Source: World Bank

The US fertility rate reached an all-time low in 2023.

Number of births per 1,000 women ages 15-44, 1909–2023



Source: [Centers for Disease Control and Prevention](#) • [Get the data](#) • [Embed](#) • [Download image](#) • [Download SVG](#)

FERTILITY & REPRODUCTIVE HEALTH

Egg Freezing

Fertility & Reproductive Health

[Egg Freezing](#)[Fertility FAQ](#)[IVF Class](#)

What is Egg Freezing?

Egg freezing, or oocyte cryopreservation, is a process in which a woman's eggs (oocytes) are extracted, frozen and stored as a method to preserve reproductive potential in women of reproductive age. The first human birth from a frozen oocyte was reported in 1986. Oocyte cryopreservation has advanced greatly over the past few years, with improved process. It is no longer considered an experimental procedure in Reproductive Medicine. The techniques leading to enhanced gamete survival allow women a much greater degree of autonomy than was

Sexual Development

- **Production of Gametes and Fertilization**
- **Development of the Sex Organs**
- **Sexual Maturation**

Sexual Development Terminology

What is the difference between sex and gender?

Sexual Development Terminology

Sex: refers to the genetic or physiological characteristics of males and females

Gender: refers to the socially influenced identity, roles, and/or behavior of an individual

Intersex: broad term describing a variety of combinations of biologically male and female characteristics

Transgender: describes a gender identity that does not necessarily correspond to biological sex at birth

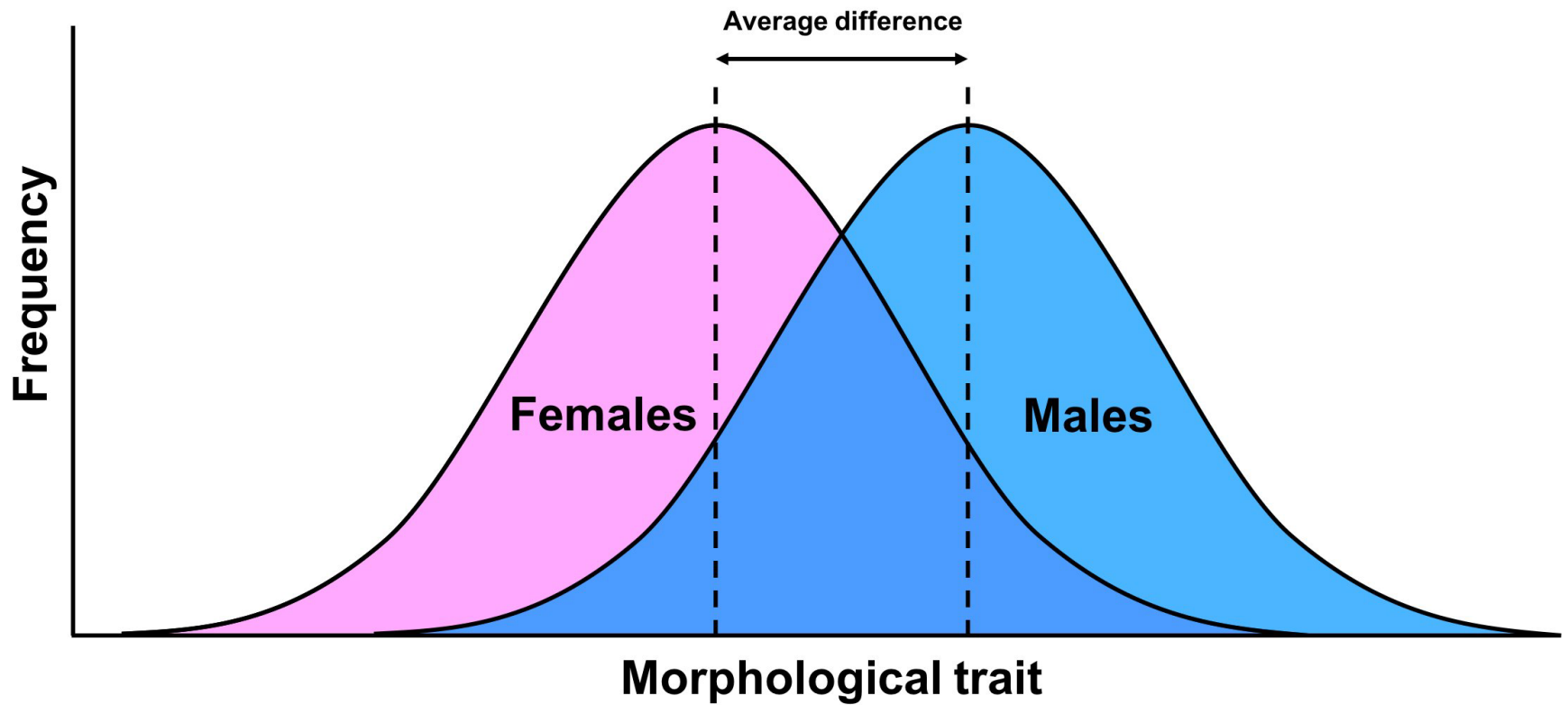
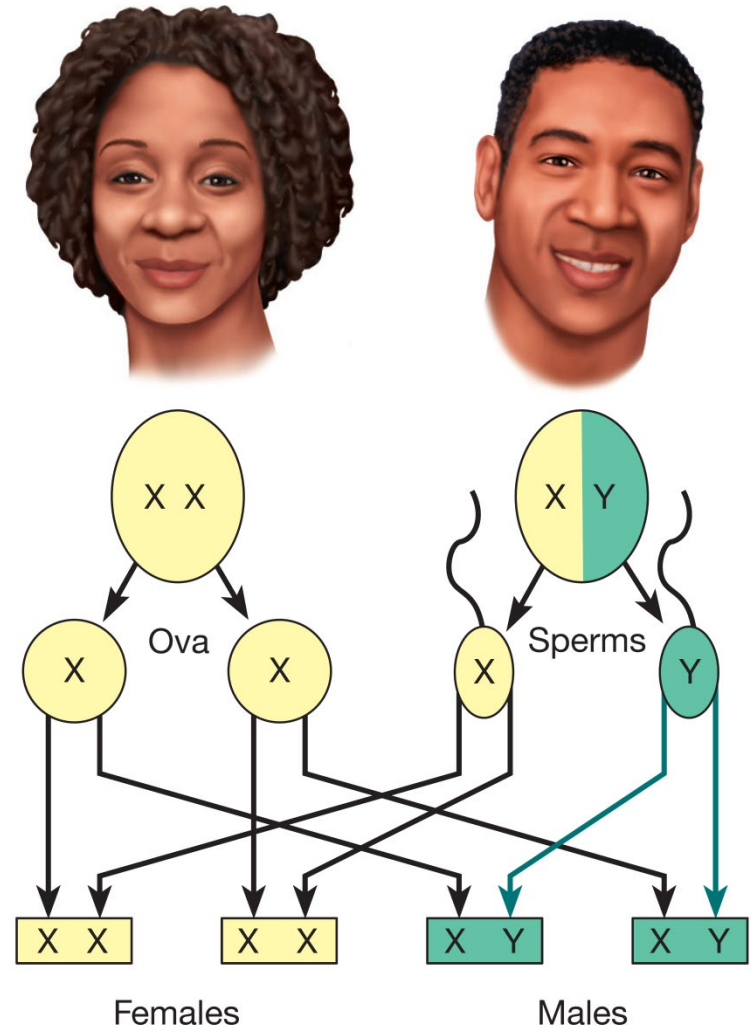


Figure 10.1

Determination of Genetic Sex



Genetic sex is determined by whether the sperm cell that fertilizes the ovum carries an X or Y chromosome. Sperm that carry an X chromosome produce genetic females; Y chromosomes produce genetic males.

Development of the Sex Organs (1 of 3)

Learning Objective 10.2 Explain how gonads, internal sex organs, and external genitalia develop.

Gonads

- Testes or ovaries; first sex organs to develop
- Produce ova or sperms and secret hormones
- **SRY** gene begins testes development
 - Produces a protein that binds to the DNA of cells in undifferentiated gonads and causes them to become gonads
 - Absence of gene produces ovaries

Maternal iron deficiency causes male-to-female sex reversal in mouse embryos

<https://doi.org/10.1038/s41586-025-09063-2>

Received: 29 June 2024

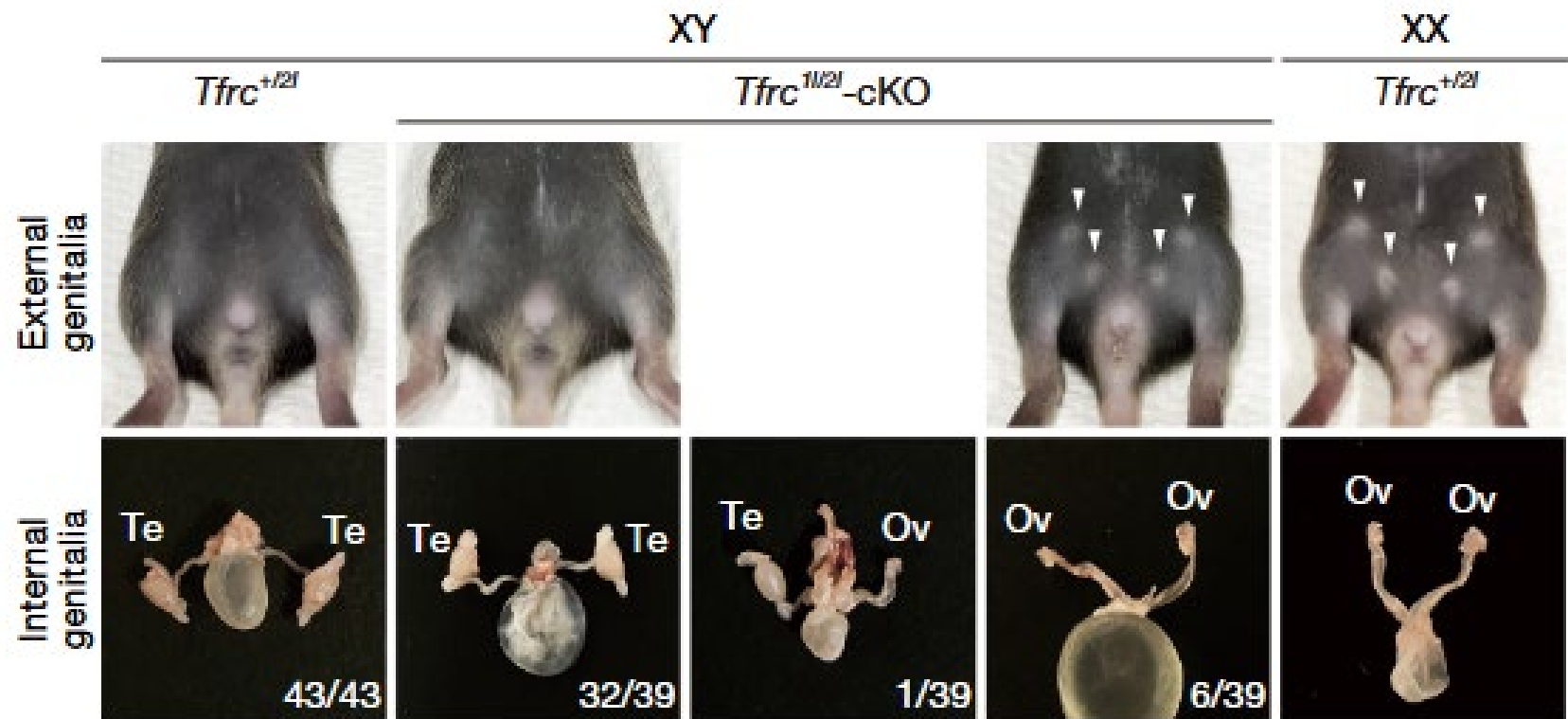
Accepted: 24 April 2025

Published online: 4 June 2025

 Check for updates

Naoki Okashita¹, Ryo Maeda¹, Shunsuke Kuroki¹, Kyona Sasaki¹, Yoko Uno¹, Peter Koopman² & Makoto Tachibana¹✉

Ferrous iron (Fe^{2+}) is essential in all eukaryotic cells for various oxidoreductase reactions, including the demethylation of DNA and proteins. Histone demethylation is required for normal epigenetic regulation of the Y-chromosomal sex-determining gene *Sry* in developing gonads during male sex determination^{1,2}. Here we investigate the potential connection between iron metabolism, histone demethylation and sex determination in mammals. We found that Fe^{2+} -producing pathways are substantially activated in mouse embryonic gonads during the sex-determining period. Chelation of iron in cultured XY gonads reduced the level of KDM3A-mediated H3K9 demethylation of *Sry*, mostly abolished *Sry* expression and caused the gonads to express ovarian markers. In vivo, conditional deletion of the gene *Tfrc*—which is required for iron incorporation—in fetal XY gonadal somatic cells, or acute pharmaceutical suppression of available iron in pregnant mice, resulted in male-to-female gonadal sex reversal in a proportion of offspring, highlighting the pivotal role of iron metabolism in male sex determination. Finally, long-term feeding of pregnant mice with a low-iron diet, when combined with a heterozygous variant of *Kdm3a* that by itself has no observable effect, suppressed *Sry* expression and caused male-to-female sex reversal in some of the progeny, revealing a connection between maternal dietary iron and fetal developmental outcomes.



What is the difference between organizational effects and activational effects?

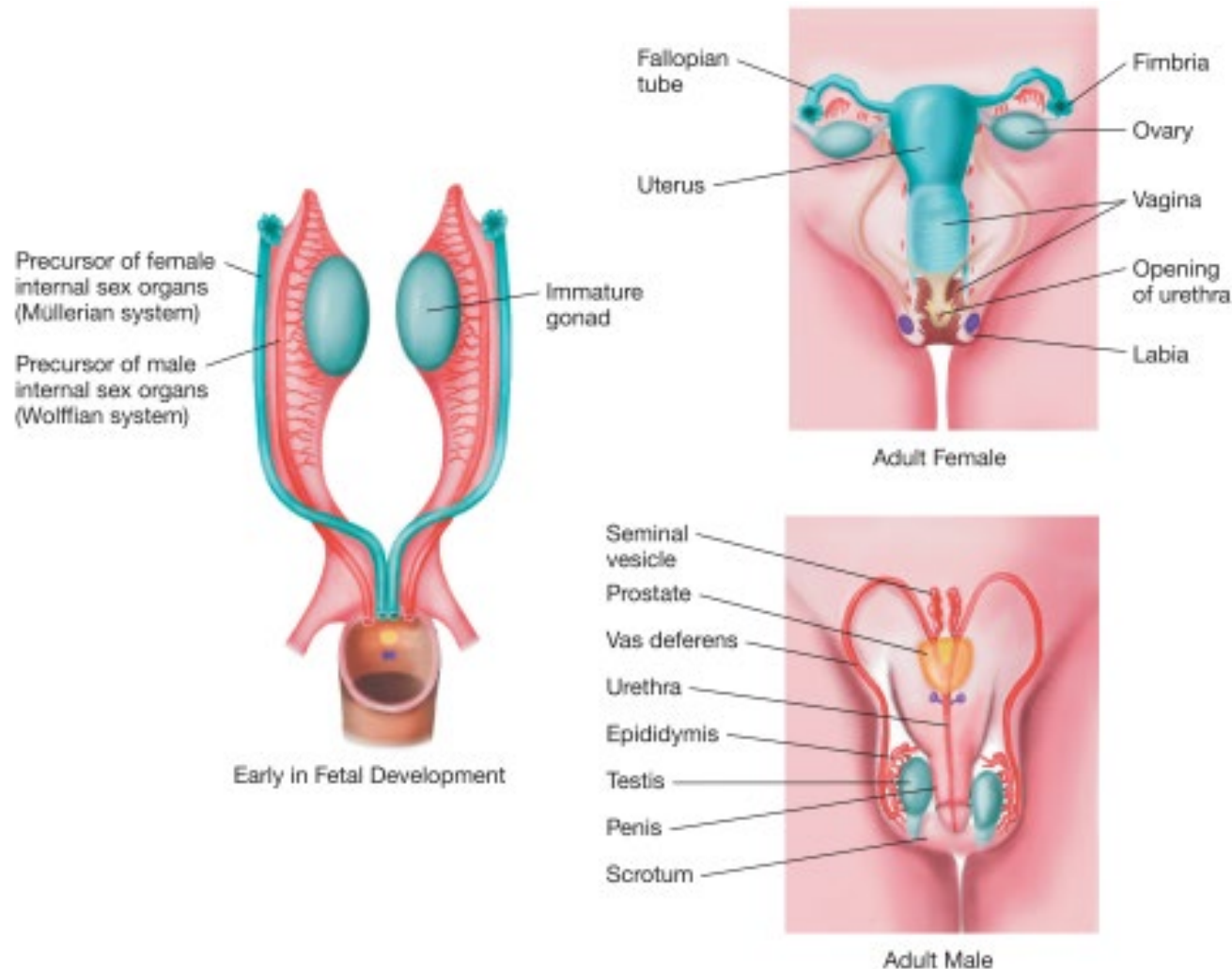
Development of the Sex Organs (1 of 3)

Learning Objective 10.2 Explain how gonads, internal sex organs, and external genitalia develop.

- **Organizational effects** of hormones influence development of sex organs and brain
- **Activational effects** occur later in life

Figure 10.2

Development of the Internal Sex Organs

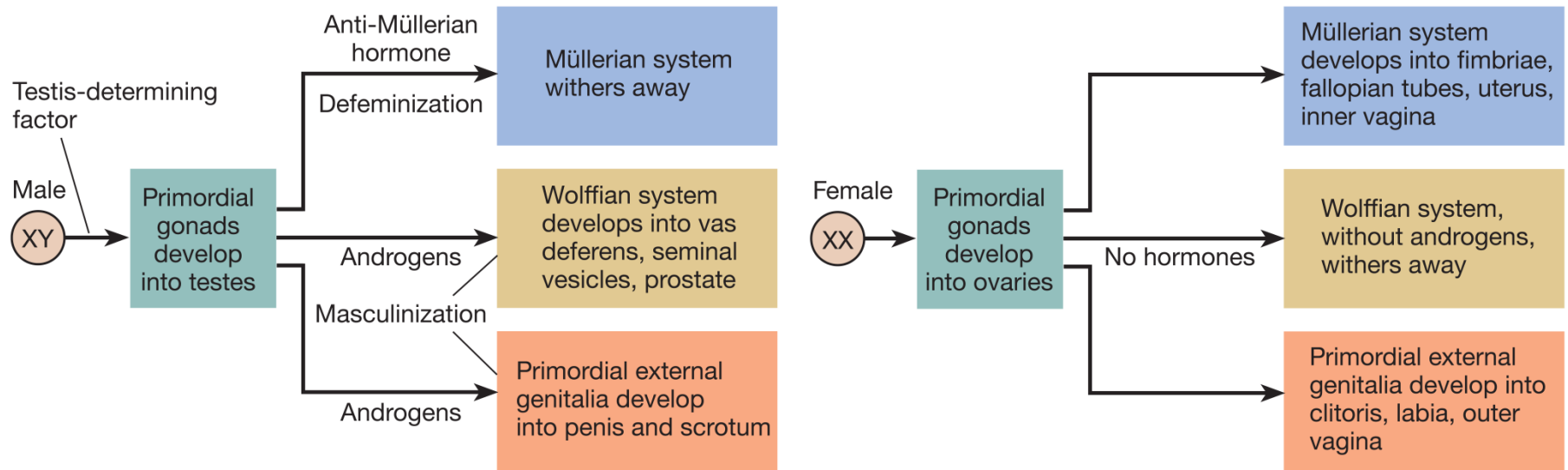


Development of the Sex Organs (2 of 3)

Internal Sex Organs

- The internal sex organs start out bisexual
- **Wolffian system** develops
 - **Androgens** (testosterone and dihydrotestosterone) and **anti-Müllerian hormone**
 - Absence leads to development of female sex organs, called the **Mullerian system**

Figure 10.4 Hormonal Control of Development of the Internal Sex Organs



Development of the Sex Organs (3 of 3)

External Genitalia

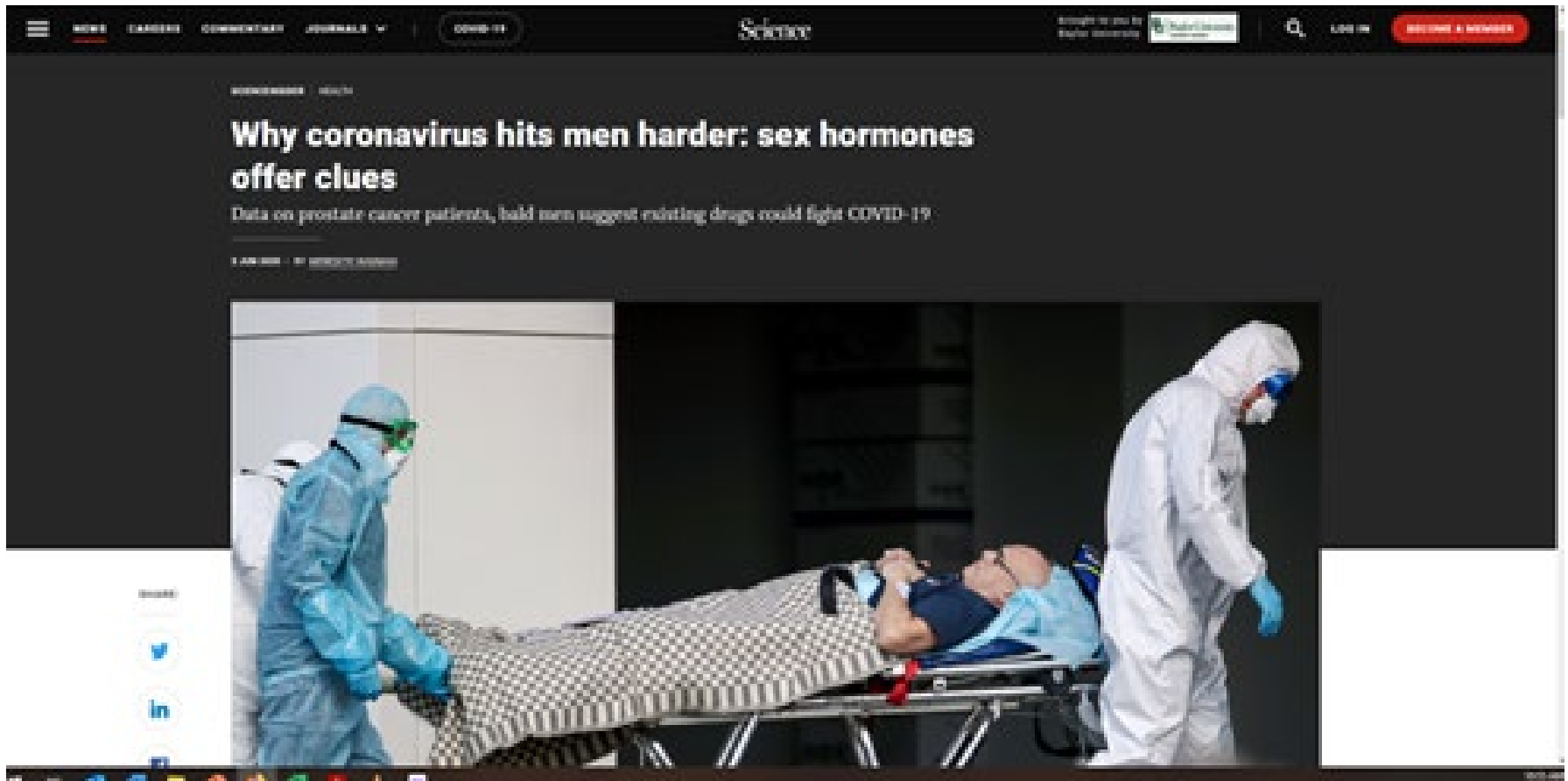
- Visible sex organs
- In presence of dihydrotestosterone, male external genitalia develops
 - Female sex hormones are not required for female genitalia development

Finasteride (Proscar and Propecia)

Inhibits the activity of the enzyme 5-alpha reductase, which is responsible for converting the male hormone testosterone to its more potent form, dihydrotestosterone (DHT).

Reduces prostate gland

Treats hair loss



Finasteride (Proscar and Propecia)

Epidemiological data from around the world have confirmed the early reports of male vulnerability.

In Lombardy in Italy, for example, men comprised 82% of 1591 patients admitted to intensive care units (ICUs) from 20 February to 18 March, according to a JAMA paper.

And male mortality exceeded that of women in every adult age group in another JAMA study of 5700 New York City patients hospitalized with COVID-19.



A protective treatment?

In one Italian study, men with prostate cancer who received drugs that suppress androgens were much less likely to be infected with SARS-CoV-2.

	Men on androgen-deprivation therapy (ADT)	Men not on ADT
Total men with prostate cancer	5273	37,161
Number infected with SARS-CoV-2	4	114
Estimated cases per 10,000	8	31

MONTOPOLI ET AL., *ANNALS OF ONCOLOGY*, [HTTPS://DOI.ORG/10.1016/J.ANNONC.2020.04.479](https://doi.org/10.1016/j.annonc.2020.04.479) (2020)

Sexual Maturation

Androgen insensitivity syndrome

Persistent Mullerian duct Syndrome

Turner Syndrome

Sexual Maturation

- Androgen insensitivity syndrome
 - Congenital lack of functioning androgen receptors
 - XY person results in development of female with testes and no internal organs

Sexual Maturation

- Persistent Mullerian duct Syndrome
 - Congenital lack of anti-mullerian hormone hormone or receptors. In male will have male and female internal organs

Sexual Maturation

- Turner Syndrome
 - Only X, lack of ovaries but normal sex organs and genitalia

Sexual Maturation (1 of 2)

Learning Objective 10.3 Compare the organizational effects of hormones on primary sex characteristics and the activational effects on secondary sex characteristics.

Primary Sex Characteristics

- Influenced by organizational effects of hormones
- Present at birth; includes gonads, internal sex organs, and external genitalia

Secondary Sex Characteristics

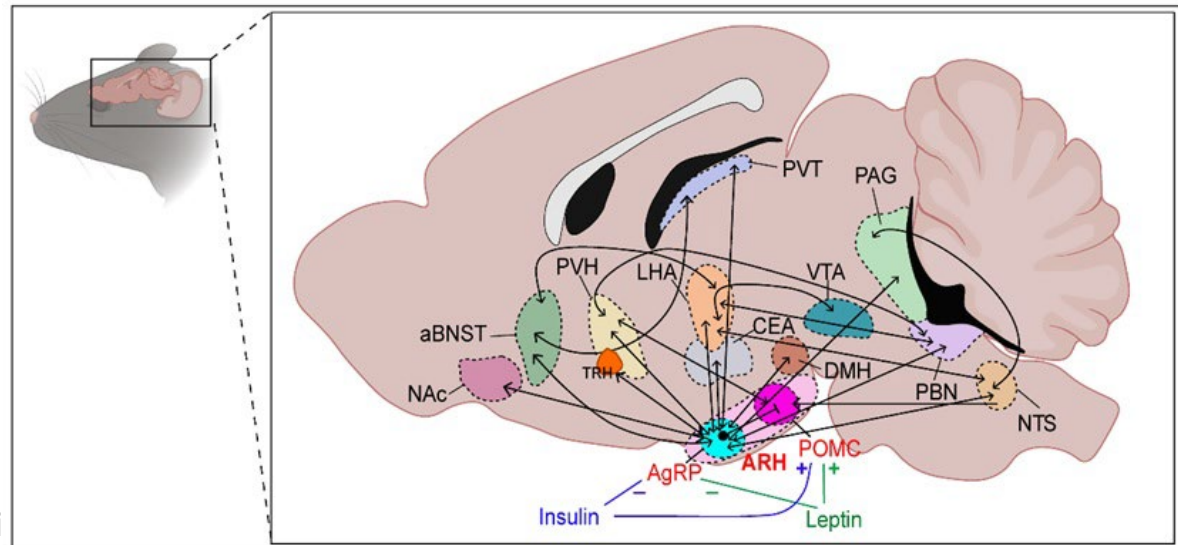
- Influenced by activational effects of hormones
- Develop during puberty

Kisspeptin

It is a peptide that controls the secretion of Gonadotropin-releasing hormone during puberty.

Kisspeptin is produced by neurons in the arcuate nucleus of the hypothalamus.

Essential for the initiation of puberty and the maintenance of male and female reproductive ability.



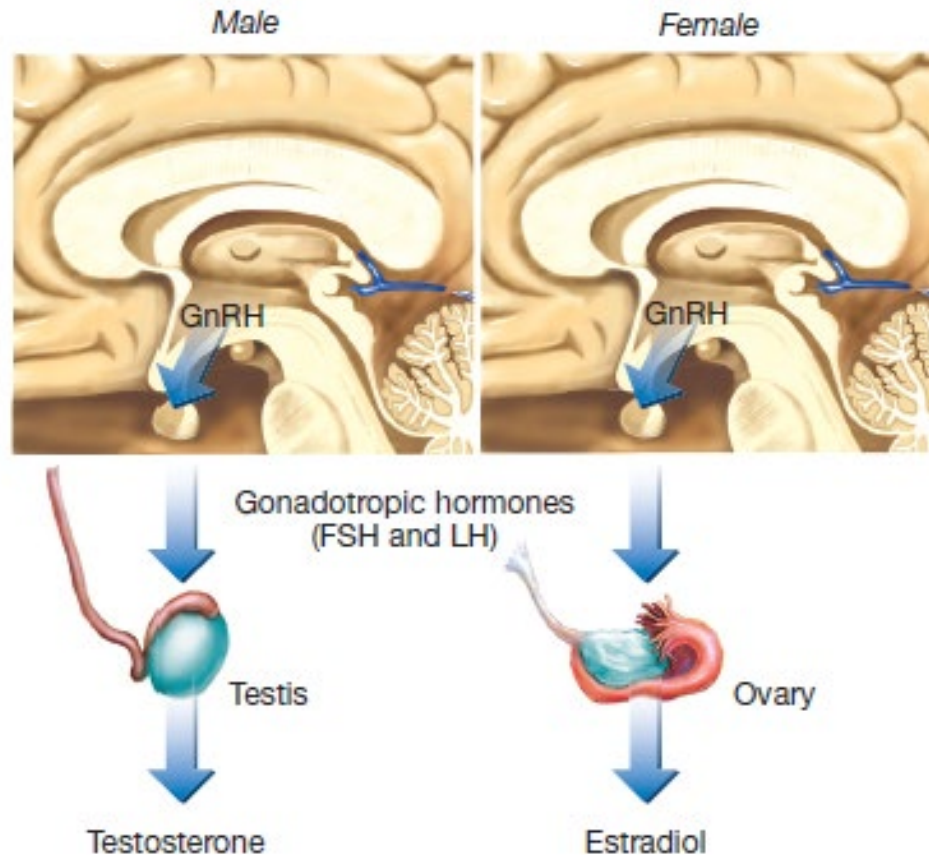
Sexual Maturation (2 of 2)

Secondary Sex Characteristics continued

- Puberty triggers hypothalamus to secrete **gonadotropin-releasing hormones (GnRH)**
 - Follicle stimulating hormone (FSH)
 - Luteinizing hormone (LH)
- Some flexibility in expression of secondary sex characteristics continues throughout life

Figure 10.5

Sexual Maturation



Puberty is initiated when the hypothalamus secretes gonadotropin-releasing hormones, which activate the secretion of gonadotropic hormones by the anterior pituitary gland.

Hormones in men

- luteinizing hormone then travels to your gonads (testicles or ovaries) and stimulates the production and release of testosterone.
- Your pituitary also releases follicle-stimulating hormone (FSH) to cause sperm production.

Hormonal Control of Sexual Behavior

- **Hormonal Control of Female Reproductive Cycles**
- **Hormonal Control of Sexual Behavior of Laboratory Animals**
- **Organizational Effects of Androgens on Behavior: Masculinization and Defeminization**
- **Human Sexual Behavior**
- **Effects of Pheromones**

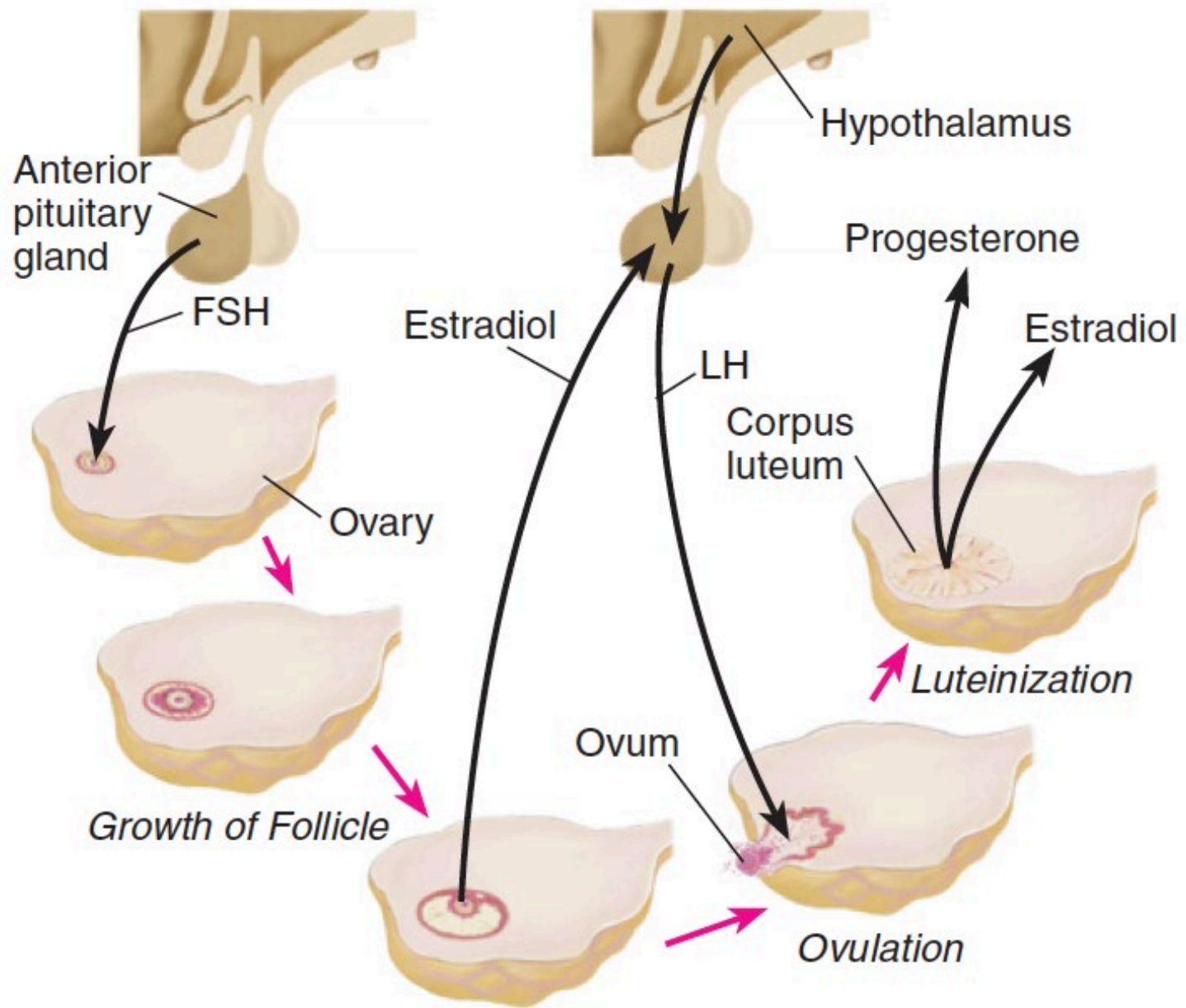
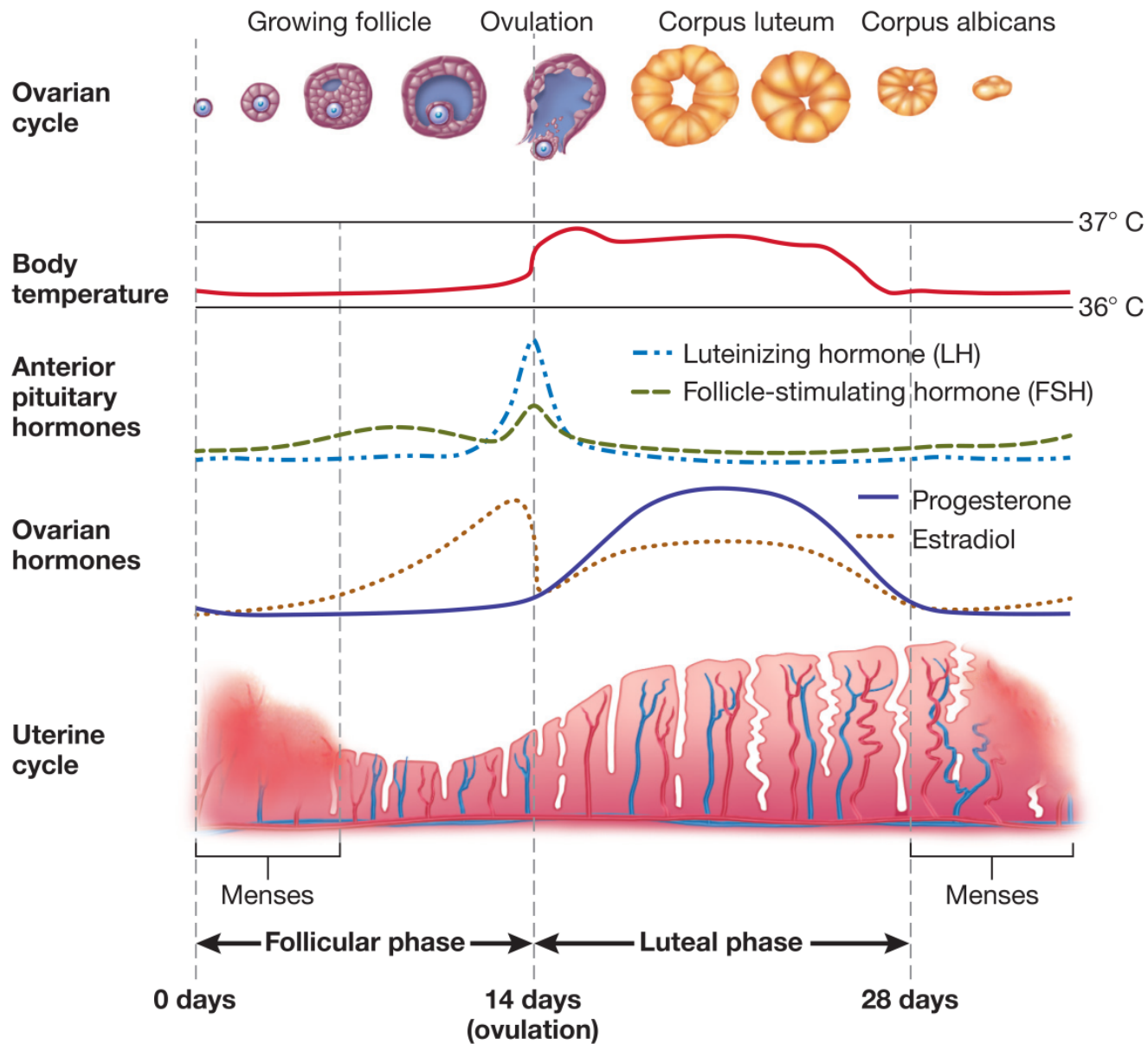


FIGURE 9.6 Neuroendocrine Control of the Menstrual Cycle.



Hormonal Control of Female Reproductive Cycles

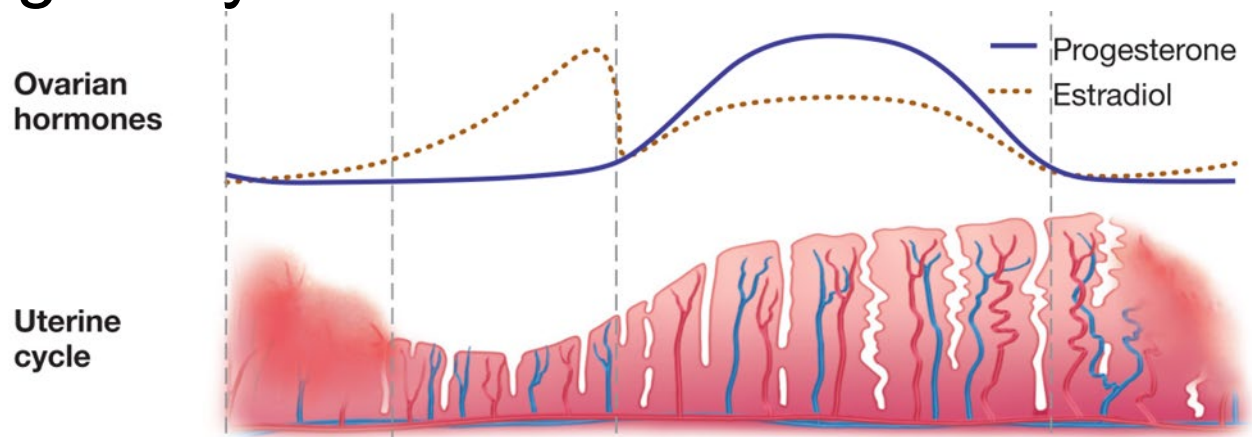
Learning Objective 10.4 Summarize the roles of hormones in phases of the menstrual cycle.

Menstrual Cycle

- Begins with secretion of gonadotropins by the anterior pituitary gland, which stimulates the growth of **ovarian follicles**
- As follicles mature, they secrete estradiol, causing lining of uterus to develop
- When estradiol reaches a critical level, it causes pituitary gland to release LH, triggering ovulation
- If the ovum is not fertilized, the uterine lining is shed

Abortion procedures - medication

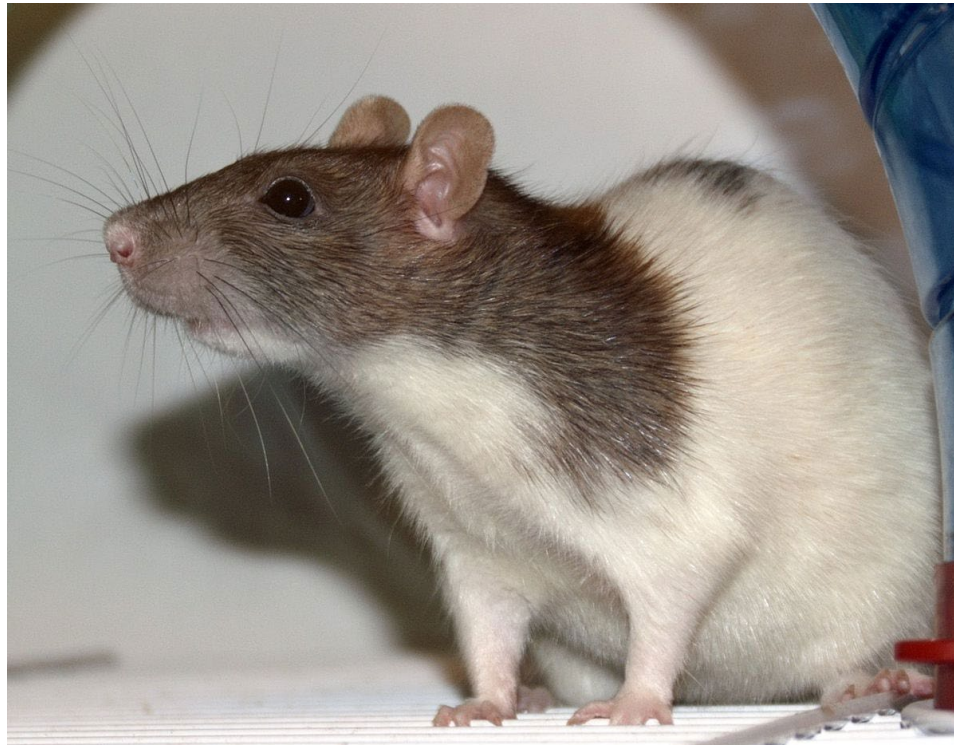
- Mifepristone (previously known as RU486) is taken by mouth. It ends a pregnancy by blocking the action of the hormone (progesterone) that supports the pregnancy.
- Misoprostol is also taken by mouth. It causes the cervix to soften and the uterus to contract to expel the pregnancy.



Which animals do we use to study sexual behavior and why?

Hormonal Control of Sexual Behavior of Laboratory Animals

Males



Hormonal Control of Sexual Behavior of Laboratory Animals

Males

- Sexual behavior of rats has been studied more than that of any or laboratory animal (Hull and Dominguez 2007)
- When male rat encounters a receptive female, he will spend some time nuzzling her and sniffing and licking her genitals, mount her, and engage in pelvic thrusting
- He will mount her several times, achieving intromission on most mountings

Hormonal Control of Sexual Behavior of Laboratory Animals

Males

- After eight to fifteen intromissions approximately 1 minute apart (each lasting only about one-quarter of a second), male will ejaculate
- After ejaculating, male refrains from sexual activity for period of time (minutes, in rat)
- Most mammals will return to copulate several times, finally showing longer pause, called refractory period

Hormonal Control of Sexual Behavior of Laboratory Animals

Refractory
period
(ree frak
to ree)

Period of time after particular action (for example, an ejaculation by a male) during which that action cannot occur again

Coolidge
effect

Restorative effect of introducing new female sex partner to male that has apparently become “exhausted” by sexual activity

Hormonal Control of Sexual Behavior of Laboratory Animals: females

Females

-

Lordosis



Hormonal Control of Sexual Behavior of Laboratory Animals

Sequence of estradiol followed by progesterone has three effects on female rats

- Receptivity
- Proceptivity
- Attractiveness

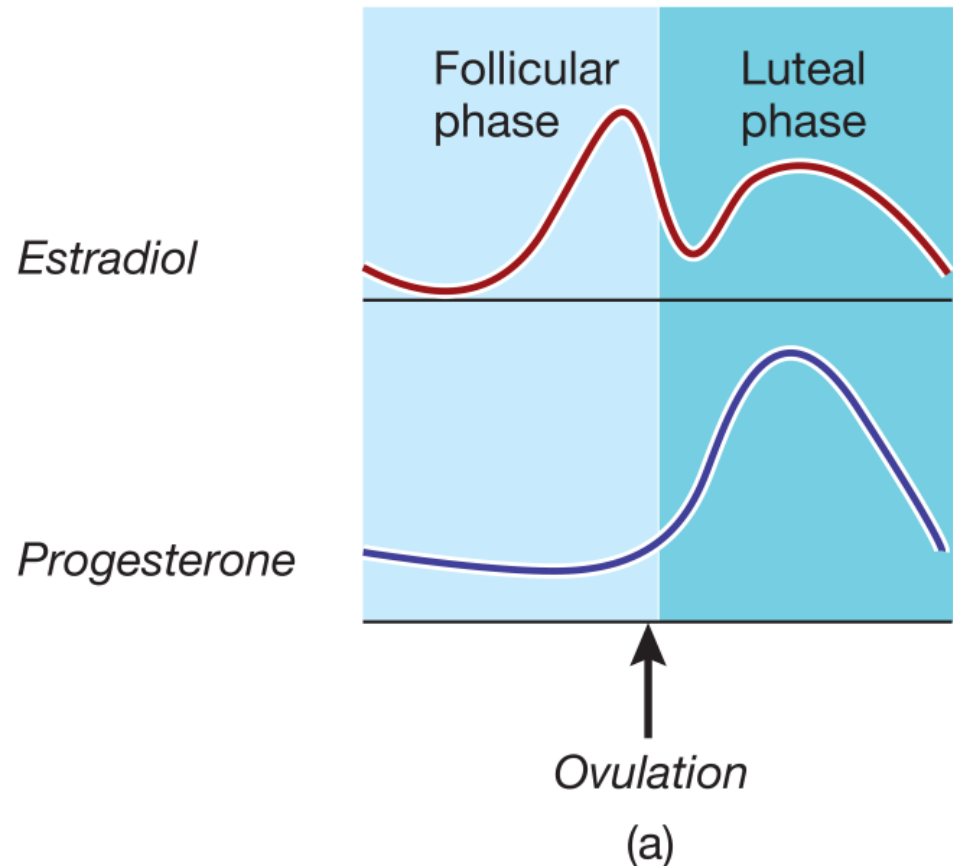
Hormonal Control of Sexual Behavior of Laboratory Animals (2 of 2)

Female Mammals

- Sexual behavior depends on estradiol and progesterone (present during estrus)
 - Estradiol and progesterone levels increase just before onset of the receptive period
 - Levels decrease after ovulation
- Oxytocin is released at the time of orgasm in both males and females

Figure 10.7 Comparison of Menstrual Cycle of Humans and Estrous Cycle of Rats (1 of 2)

Human menstrual cycle



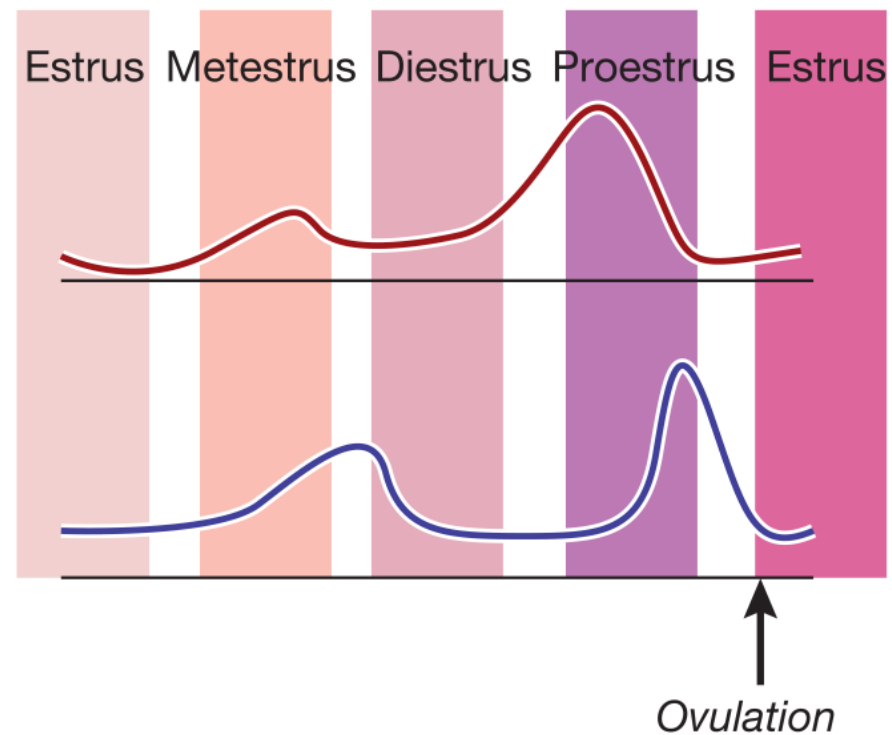
(a) The human menstrual cycle consists of two phases (follicular and luteal), which are each approximately 14 days long. Ovulation occurs at the midpoint of the cycle.

Source: (Becker JB, Arnold AP, Berkley KJ, Blaustein JD, Eckel LA, Hampson E, Herman JP, Marts S, Sadée W, Steiner M, Taylor J, Young E: Strategies and methods for research on sex differences in brain and behavior. *Endocrinology*. 2005, 146: 1650–1673.)

Figure 10.7 Comparison of Menstrual Cycle of Humans and Estrous Cycle of Rats (2 of 2)

(b) The rat estrous cycle consists of four phases which are approximately one day long. Ovulation occurs during the estrus phase. Notice that the estrus phase is shown twice in the figure to demonstrate the cyclic pattern of the phases. Source: (Becker JB, Arnold AP, Berkley KJ, Blaustein JD, Eckel LA, Hampson E, Herman JP, Marts S, Sadee W, Steiner M, Taylor J, Young E: Strategies and methods for research on sex differences in brain and behavior. Endocrinology. 2005, 146: 1650–1673.)

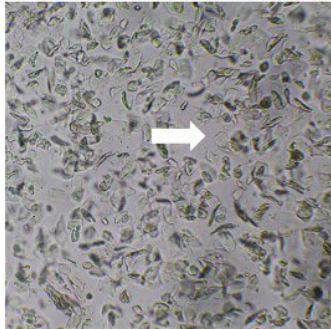
Rat estrous cycle



(b)

Rat Estrus Phase

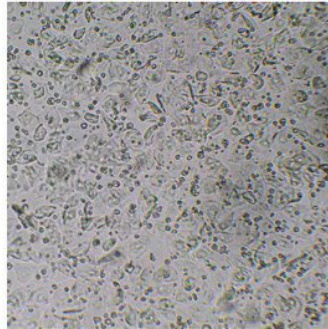
Proestrus



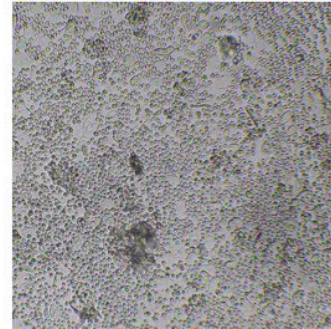
Estrus



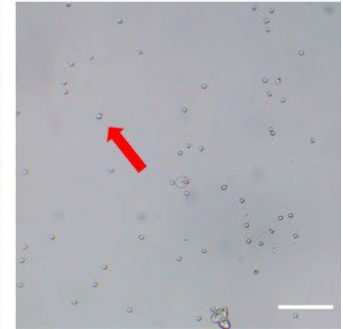
Metestrus



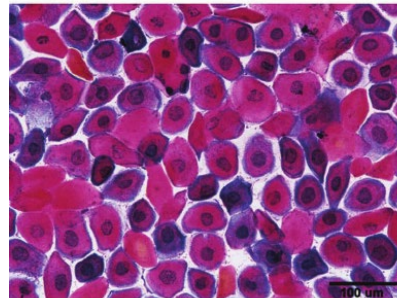
Diestrus



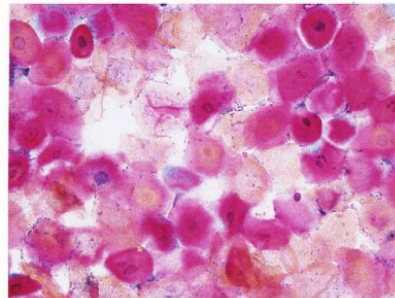
Few



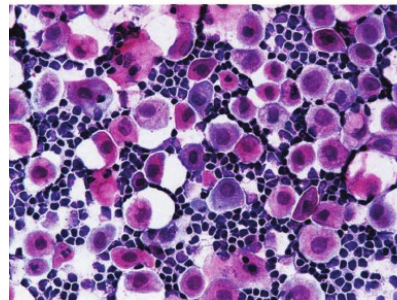
Proestrus



Estrus



Metestrus



Diestrus

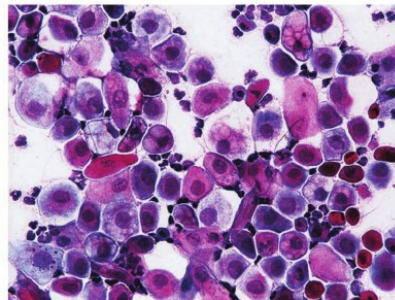


Figure 10.8

Organizational Effects of Testosterone

Hormone Treatment

Immediately after birth	When rat is fully grown	Resulting Sexual Behavior	
None	E + P	Female: yes	Male: no
None	Testosterone	Female: no	Male: no
Activational effect of estradiol and progesterone in nonandrogenized animal			
Testosterone	E + P	Female: no	Male: no
Testosterone	Testosterone	Female: no	Male: yes
Evidence of <i>defeminization</i>: Estradiol + progesterone fails to facilitate female sexual behavior			
		Evidence of <i>masculinization</i>: Testosterone facilitates male sexual behavior	

Around the time of birth, testosterone masculinizes and defeminizes rodents' sexual behavior.

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ENDOCRINE-DISRUPTING CHEMICALS: HAZARDS AND OPPORTUNITIES

Overview

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Endocrine-Disrupting Chemicals: Hazards and Opportunities (ECHO)

The course focuses on the chemical, biological, and societal challenges of endocrine-disrupting chemicals (EDCs) and new opportunities for moving forward.

Course/Program Dates: May 17, 2024 - May 31, 2024

Application due date: Jan 15, 2024

 [APPLY FOR THIS COURSE](#)

Directors: [Patricia Hunt](#), Washington State University; [Joan Ruderman](#), Princeton University

Organizational Effects of Androgens on Behavior: Masculinization and Defeminization

Learning Objective 10.6 Contrast behavioral masculinization and defeminization in rodents.

Masculinization and Defeminization

- Suppressing or stimulating development of neural circuits controlling sexual behavior leads to behavioral defeminization or behavioral masculinization

Human Sexual Behavior (1 of 2)

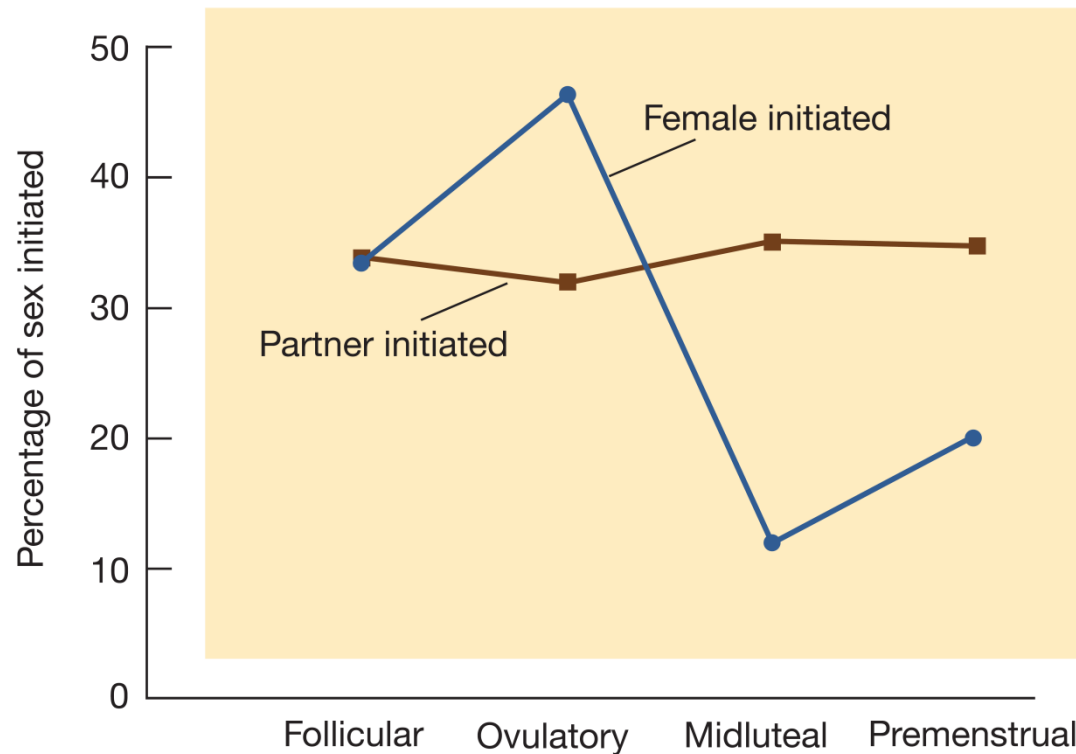
Learning Objective 10.7 Compare the activational effects of hormones on sexual behavior of human males and females.

Activational Effects of Sex Hormones in Women

- Ovarian hormones influence a women's sexual interest, behavior and physiology
 - Studies suggest female sexual interest can also be stimulated by androgens

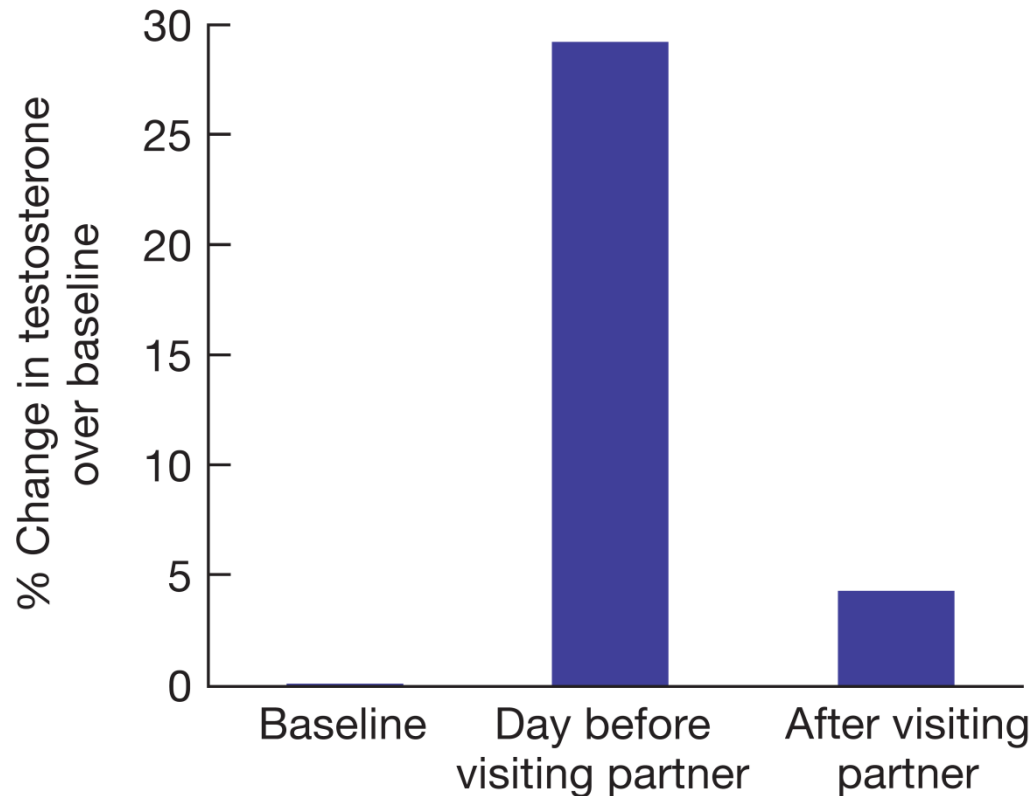
Figure 10.9

Sexual Activity of Heterosexual Couples



This graph shows the distribution of sexual activity initiated by the man and by the woman during the phases of the woman's menstrual cycle. Source: (Wallen, K. (2001). Sex and context: hormones and primate sexual motivation. *Hormones and behavior*, 40(2), 339–357.)

Figure 10.10 Long-Distance Relationships and Testosterone Levels in Women



Source: (Data from Hamilton, L. D., & Meston, C. M. (2010). The effects of partner togetherness on salivary testosterone in women in long distance relationships. *Hormones and behavior*, 57(2), 198–202.)

Human Sexual Behavior (2 of 2)

Activational Effects of Sex Hormones in Men

- Testosterone has an activational effect on the sexual behavior of men
 - Sperm production and sexual interest ceases without testosterone
- Testosterone is also affected by sexual activity or by thinking about it

Why Animals Talk



*The
New Science
of Animal
Communication*

ARIK KERSHENBAUM

Effects of Pheromones

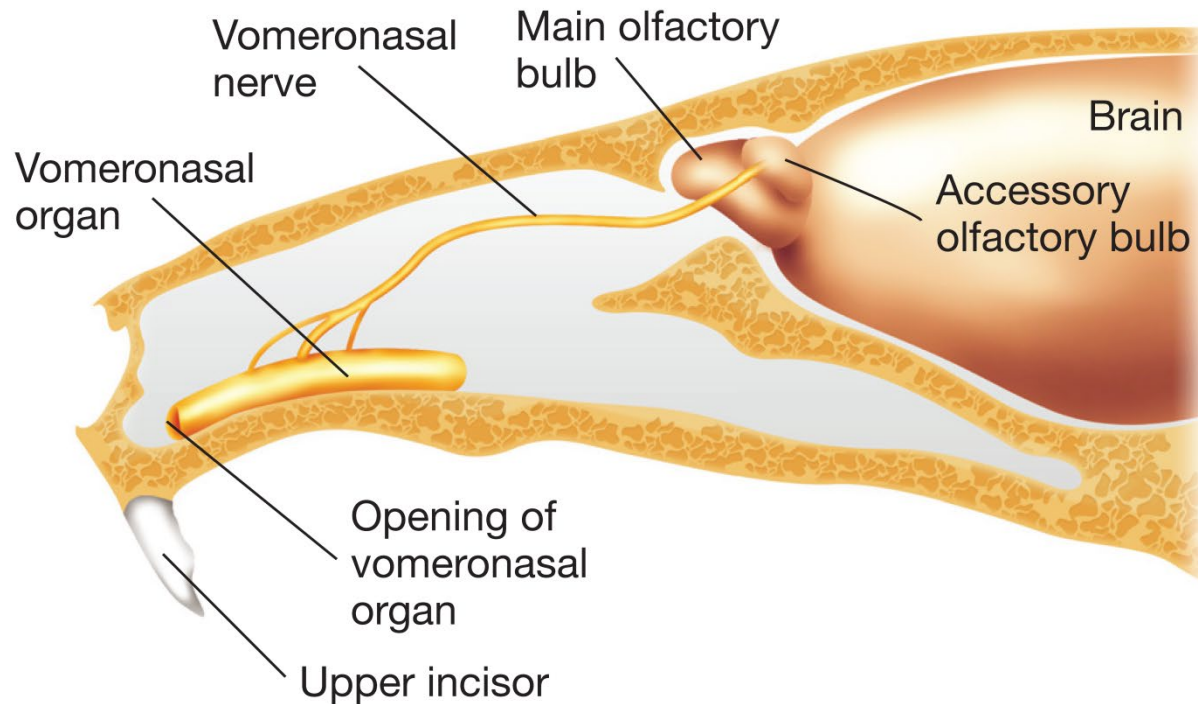
Learning Objective 10.8 Describe the roles of pheromones in reproductive behavior of humans and other animals.

Pheromones

- Carry messages from one animal to another
- In most species, detected by **vomeronasal organ** and convey information about an animal's sex, estrous condition, and identity
 - Odorants found mostly in urine and other secretions
- Pheromones play a role in human behavior, and can influence menstrual cycles, mood, and changes in attractiveness ratings and brain activity

Figure 10.11

The Rodent Accessory Olfactory Bulb



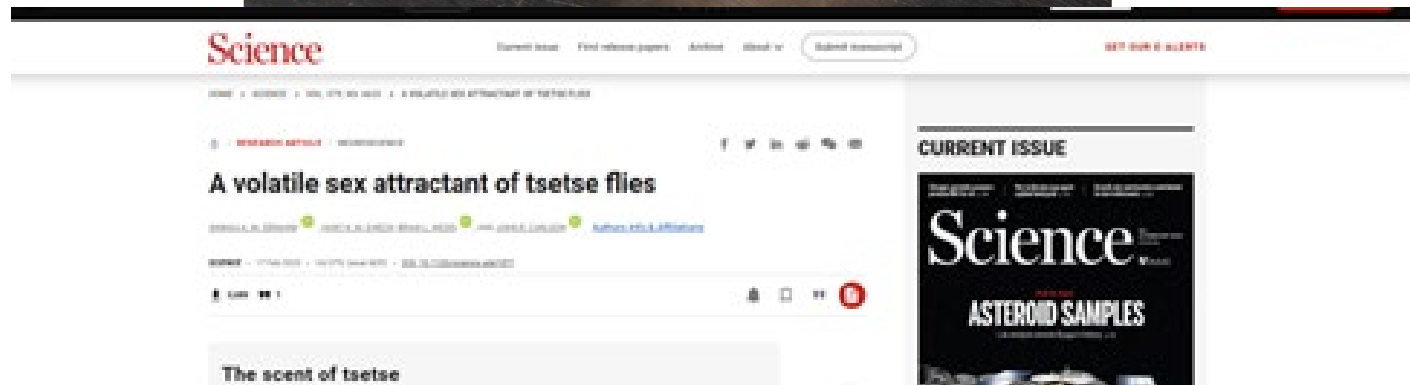
Source: (Based from Wysocki, C. J. (1979). Neurobehavioral evidence for the involvement of the vomeronasal system in mammalian reproduction. *Neuroscience & Biobehavioral Reviews*, 3(4), 301–341.)

Table 10.3 Pheromones and Behavior

Effect of Pheromone	Description
Lee-Boot Effect	Estrous cycles stop in groups of female rodents living together
Whitten Effect	Groups of female rodents who are not cycling are exposed to male urine and begin cycling synchronously
Vandenbergh Effect	Acceleration of onset of puberty in female rodents exposed to odor of male
Bruce Effect	Exposure to urine of novel male results in failure of pregnancy in newly pregnant female

The perfume of the tsetse fly opens new doors to disease control

By Amy McDermott | February 24, 2023



Effects of Pheromones

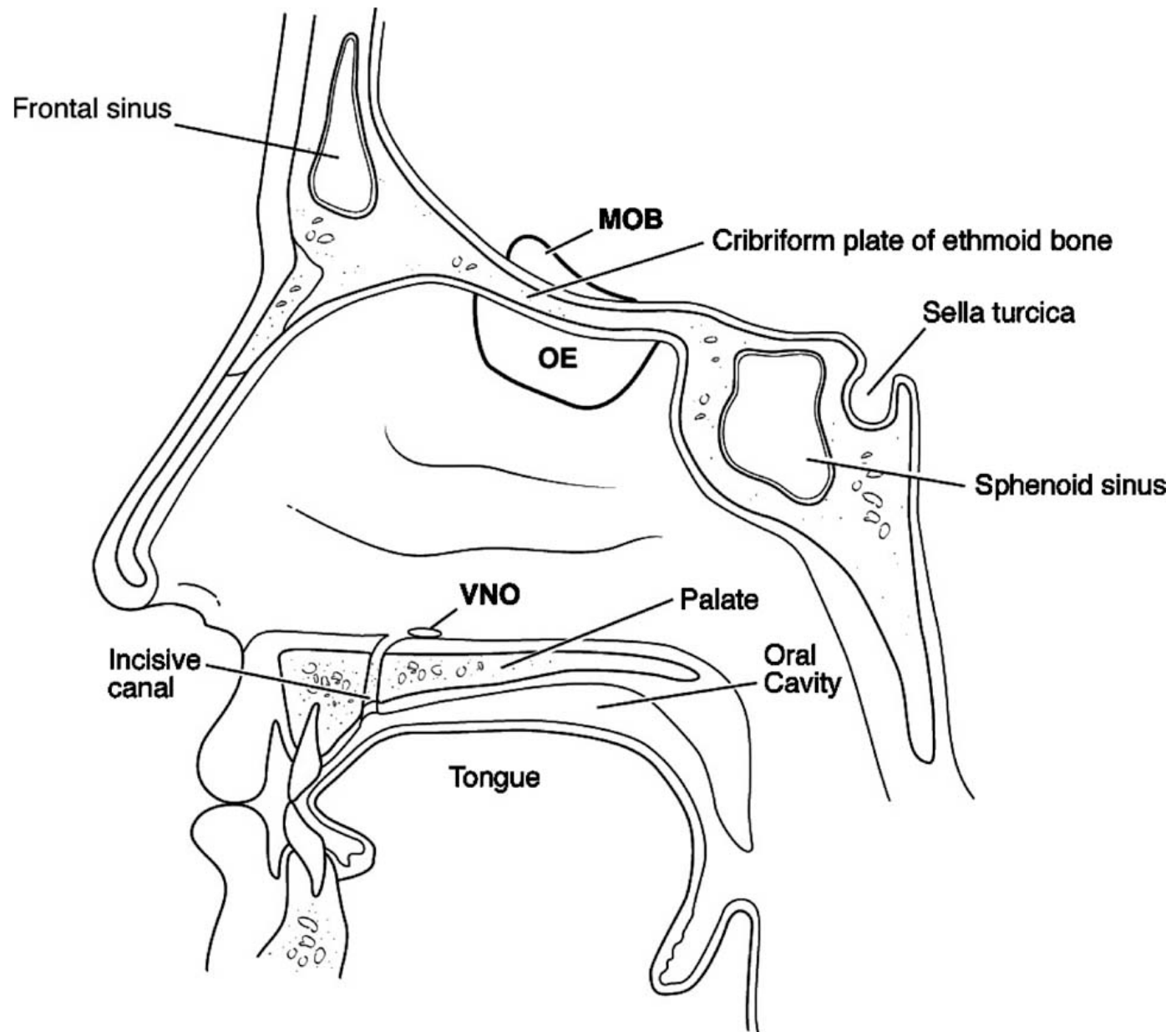
Learning Objective 10.8 Describe the roles of pheromones in reproductive behavior of humans and other animals.

New control methods for the biting flies, which have a long history of spreading devastating diseases across Sub-Saharan Africa.



Tsetse traps, such as this one set up along the Kafu River near Kafu, Uganda, are often baited with the odor of cow urine, which is distilled into a chemical attractant in a small open-topped jar or packet sitting at the trap's base. Image credit: Brian Weiss

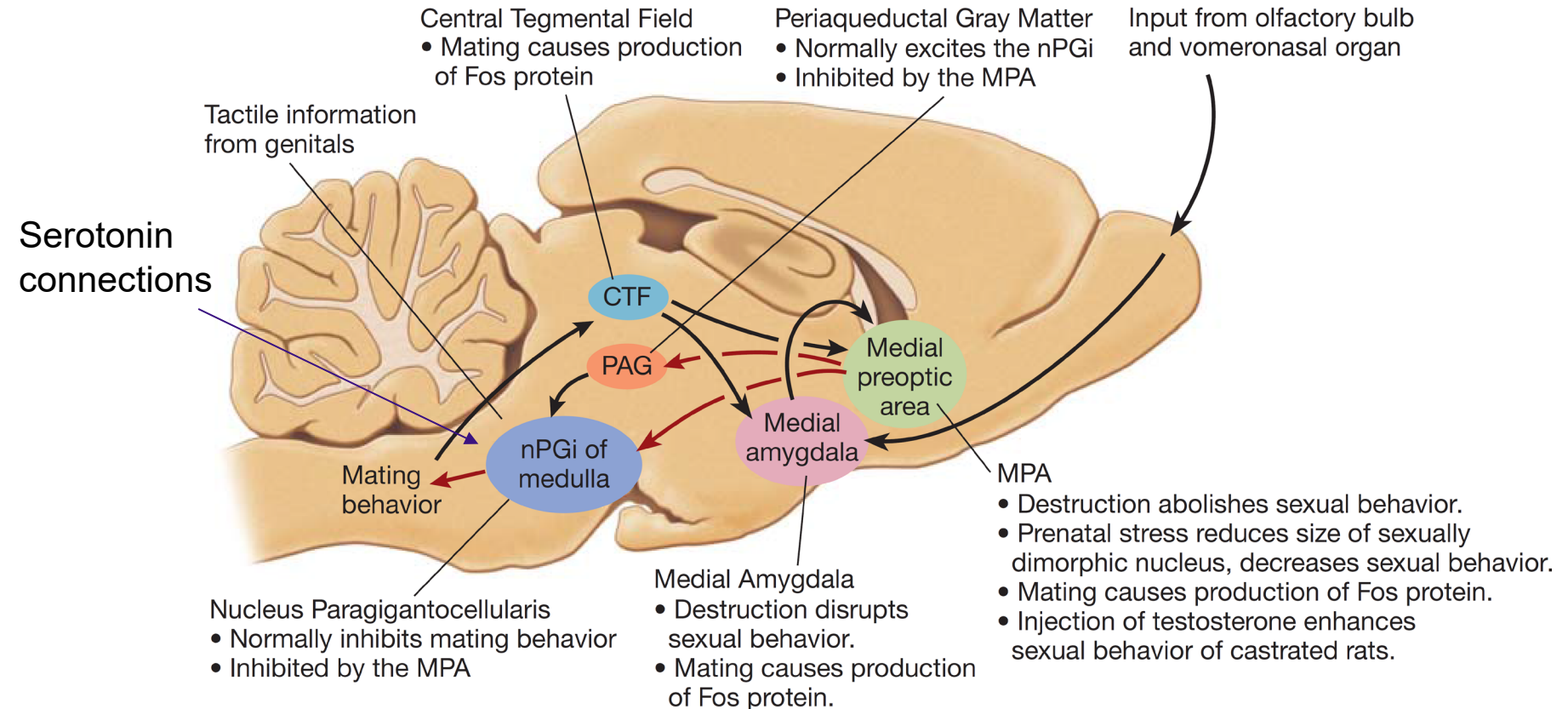




Neural Control of Sexual Behavior

- **Males**
- **Females**
- **Formation of Pair Bonds**

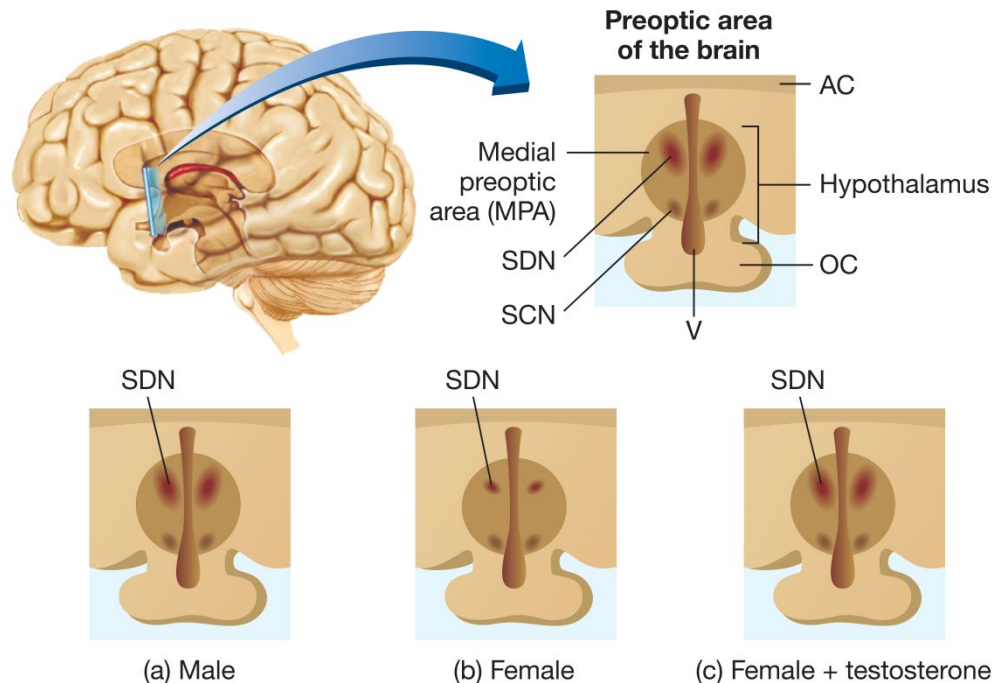
Male Sexual Behavior



This schematic shows a possible explanation of the interacting excitatory effects of pheromones, genital stimulation, and testosterone on male sexual behavior. Black arrows represent excitatory pathways. Red arrows represent inhibitory pathways.

Figure 10.12

Preoptic Area of the Rat Brain



This figure shows the preoptic area of a rat brain in (a) a typical male, (b) a typical female, and (c) an androgenized female, given an injection of testosterone shortly after birth. SDN = sexually dimorphic nucleus of the preoptic area; OC = optic chiasm; V = third ventricle; SCN = suprachiasmatic nucleus; AC = anterior commissure. Source: (Based on Gorski, R. A., in *Neuroendocrine Perspectives*, Vol. 2, edited by E. E. Müller and R. M. MacLeod. Amsterdam: Elsevier-North Holland, 1983.)

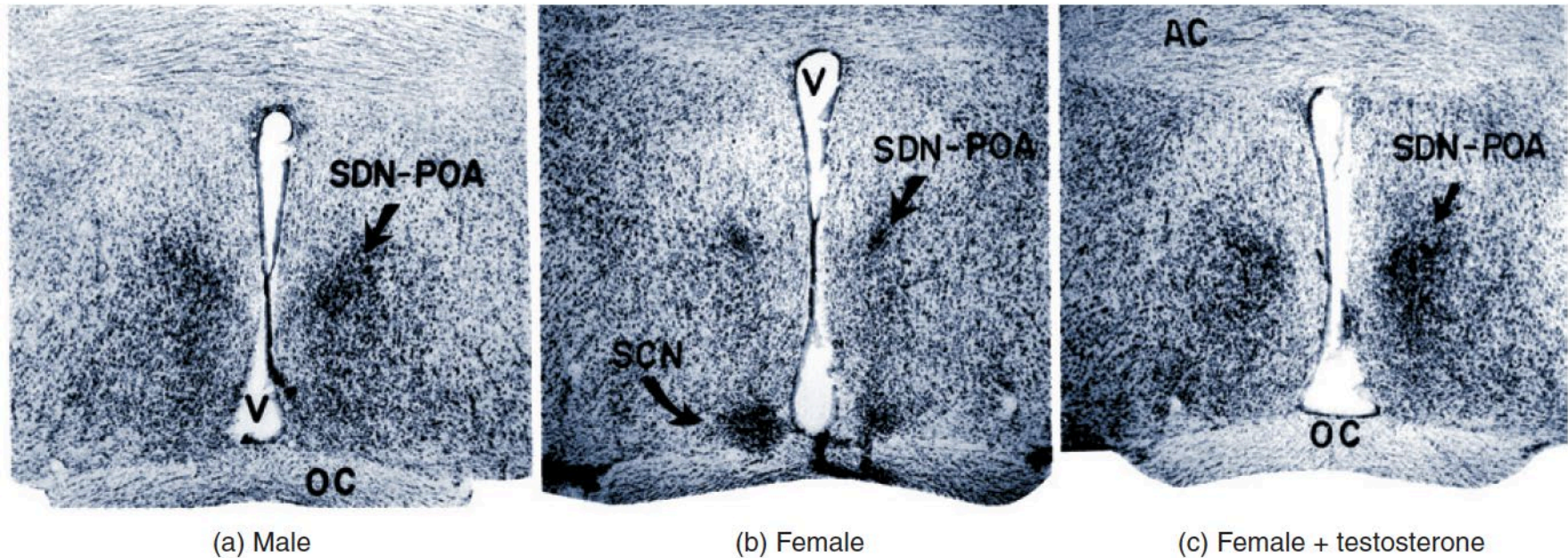


FIGURE 9.11 Preoptic Area of the Rat Brain. These photomicrographs of sections through the preoptic area of a rat brain show (a) normal male, (b) normal female, and (c) androgenized female, given an injection of testosterone shortly after birth. SDN-POA = sexually dimorphic nucleus of the preoptic area; OC = optic chiasm; V = third ventricle; SCN = suprachiasmatic nucleus; AC = anterior commissure.

Males

Sexual Behavior

- Erection, and ejaculation are controlled by neurons in the spinal cord
 - Spinal ejaculation generator
 - Brain mechanisms have both excitatory and inhibitory control

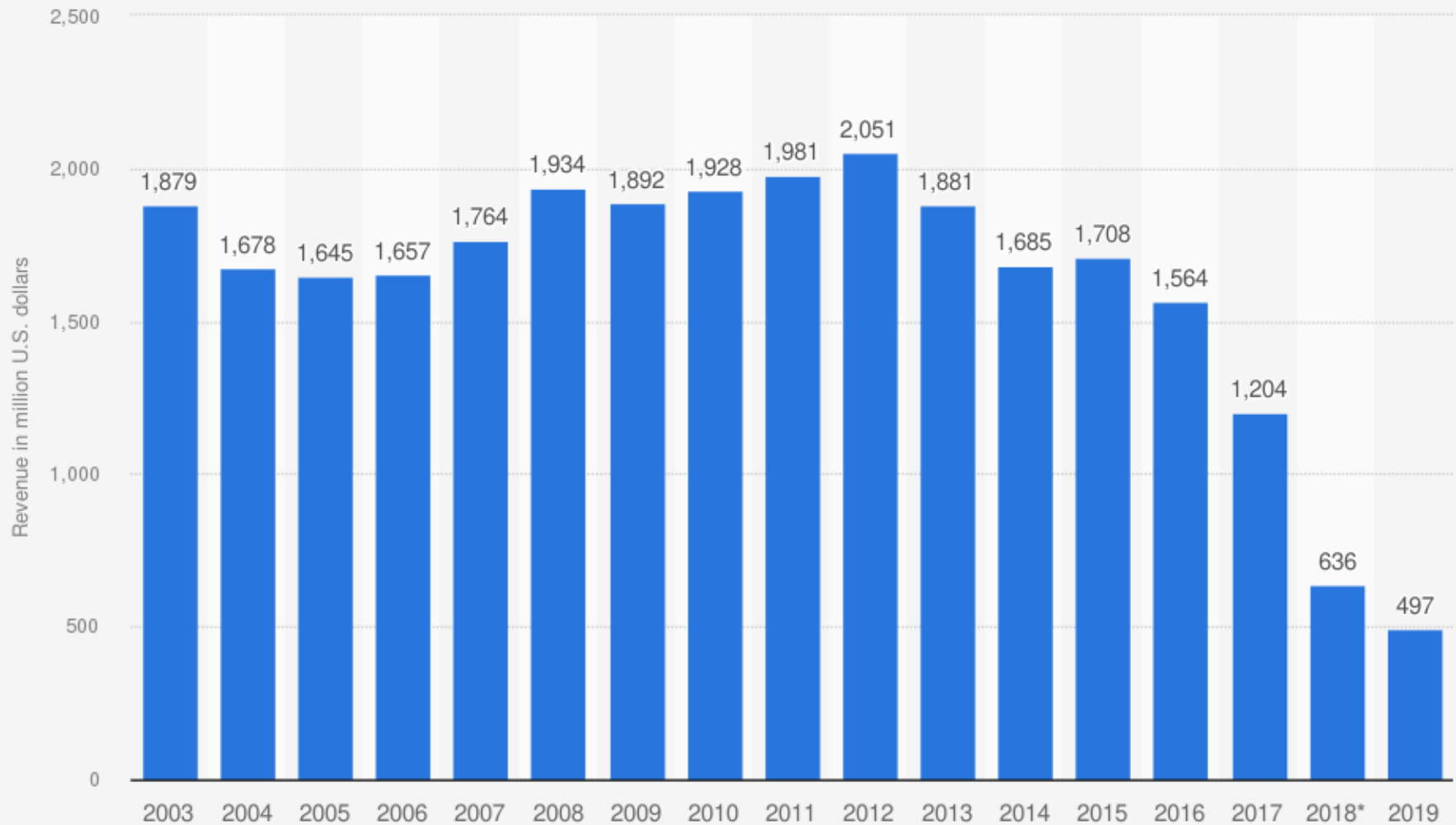
Males: Sexual Behavior

- The nPGi has inhibitory effects on spinal cord sexual reflexes, so one of the tasks of the pathway originating in the MPA is to suppress this inhibition. The MPA suppresses the nPGi directly through an inhibitory pathway and also does so indirectly by inhibiting the activity of the PAG, which normally excites the nPGi.
- Serotonin connections: Men who take SSRIs as a treatment for depression often report that they have no trouble attaining an erection but have difficulty achieving an ejaculation.

Males: Sexual Behavior Viagra

- Viagra, also known as sildenafil citrate, is a medication used to treat erectile dysfunction (ED) in men. It works by increasing blood flow to the penis, which allows a man to achieve and maintain an erection when sexually stimulated.
- Inhibits the activity of an enzyme called phosphodiesterase type 5 (PDE5). PDE5 breaks down cyclic guanosine monophosphate (cGMP), a molecule that is responsible for relaxing the smooth muscles in the blood vessels of the penis and allowing increased blood flow.
- Viagra does not work without sexual stimulation

Worldwide revenue of Pfizer's Viagra from 2003 to 2019 (in million U.S. dollars)

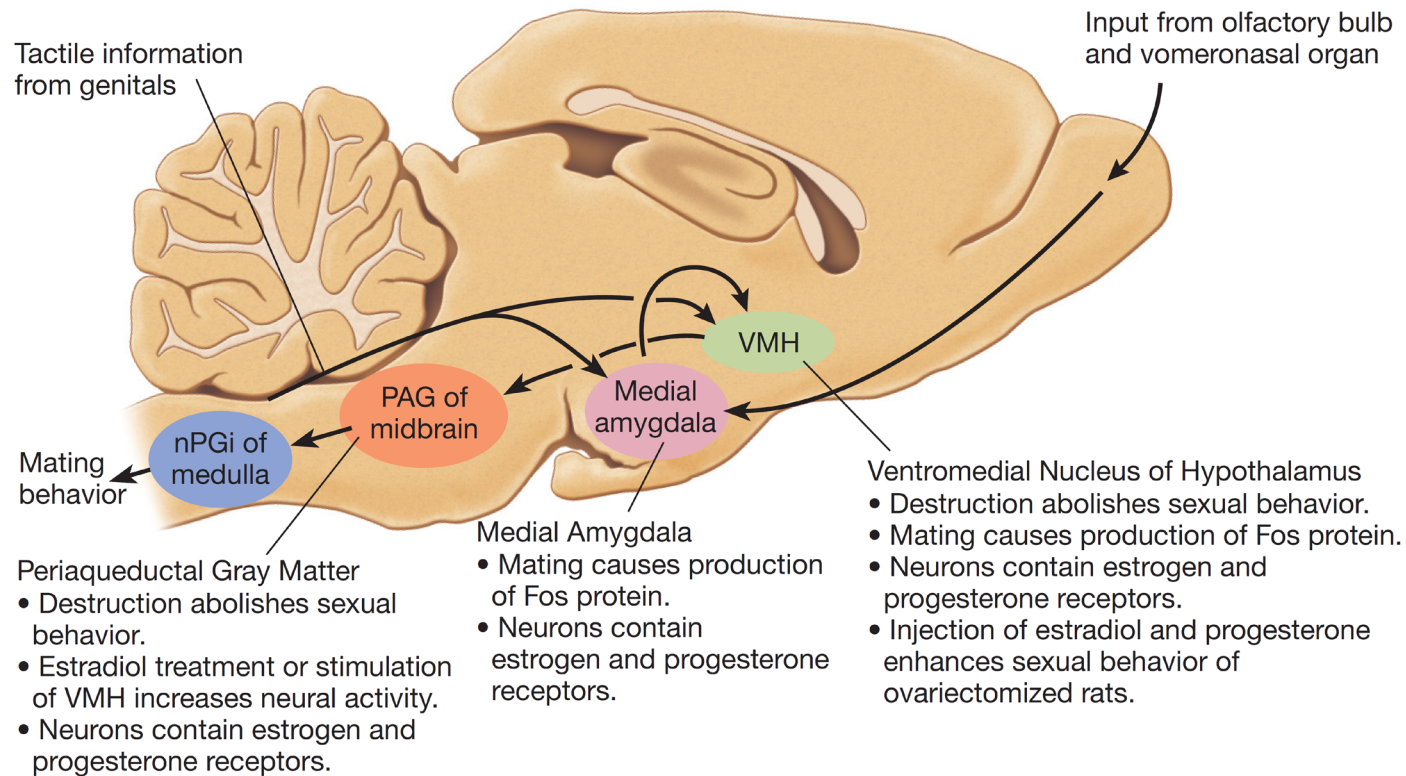


Source
Pfizer
© Statista 2023

Additional Information:
Worldwide

Figure 10.14

Female Sexual Behavior



This schematic shows a possible explanation of the interacting excitatory effects of pheromones, genital stimulation, and estradiol and progesterone on female sexual behavior. All pathways are excitatory.

Females

Learning Objective 10.10 Identify the roles of brain mechanisms in female sexual behavior.

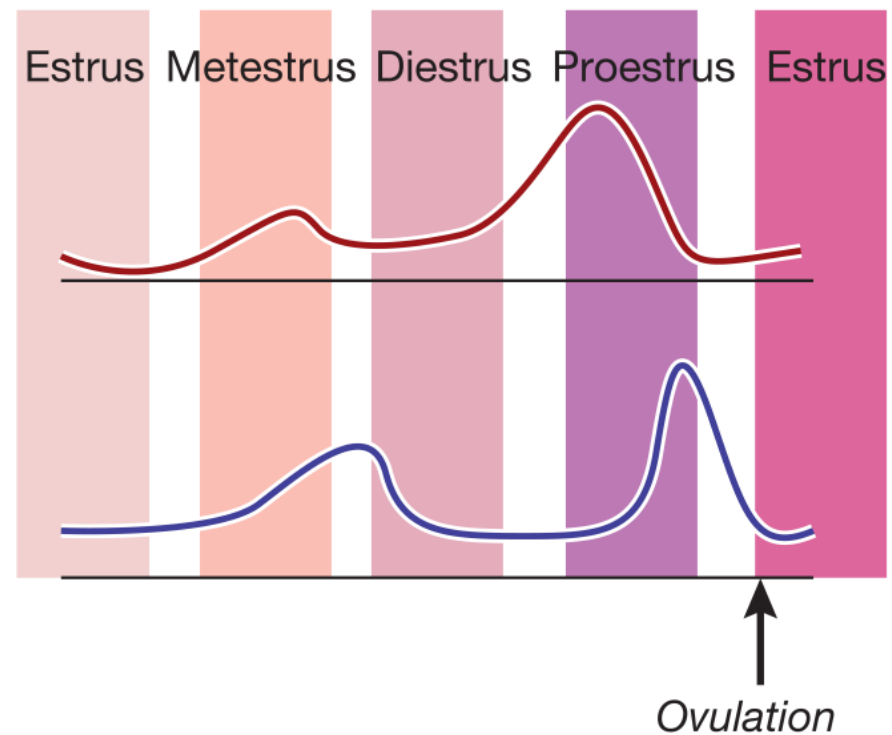
Sexual Behavior

- Stimulating the **ventromedial nucleus of the hypothalamus (VMH)** facilitates sexual behavior; destroying it abolishes it
 - Role of estradiol and progesterone

Figure 10.7 Comparison of Menstrual Cycle of Humans and Estrous Cycle of Rats (2 of 2)

(b) The rat estrous cycle consists of four phases which are approximately one day long. Ovulation occurs during the estrus phase. Notice that the estrus phase is shown twice in the figure to demonstrate the cyclic pattern of the phases. Source: (Becker JB, Arnold AP, Berkley KJ, Blaustein JD, Eckel LA, Hampson E, Herman JP, Marts S, Sadee W, Steiner M, Taylor J, Young E: Strategies and methods for research on sex differences in brain and behavior. Endocrinology. 2005, 146: 1650–1673.)

Rat estrous cycle



(b)

Formation of Pair Bonds

Learning Objective 10.11 Compare the roles of oxytocin and vasopressin in pair bonding.

Pair Bonds



Prairie Voles



Meadow Voles

Formation of Pair Bonds

Learning Objective 10.11 Compare the roles of oxytocin and vasopressin in pair bonding.

How common are pair bonds?

What are the main neuropeptides involved with pair bonding?

Formation of Pair Bonds

Learning Objective 10.11 Compare the roles of oxytocin and vasopressin in pair bonding.

Pair Bonds

- Pair bonding occurs in about 5 percent of mammals species, including all types of human couples
- Vasopressin and oxytocin peptides facilitate pair bonding
 - Male voles: vasopressin plays major role
 - Female voles: oxytocin plays major role
 - Studies show both are important in human pair bonding

Formation of Pair Bonds

Learning Objective 10.11 Compare the roles of oxytocin and vasopressin in pair bonding.

Pair Bonds Males

- Monogamous voles have a higher level of V1a vasopressin receptors
- Blocking V1a receptors diminishes pair bonding
- Injecting a genetically modified virus that contains the gene for V1a receptor into the ventral forebrain of the meadow voles transformed them into monogamous voles

Formation of Pair Bonds

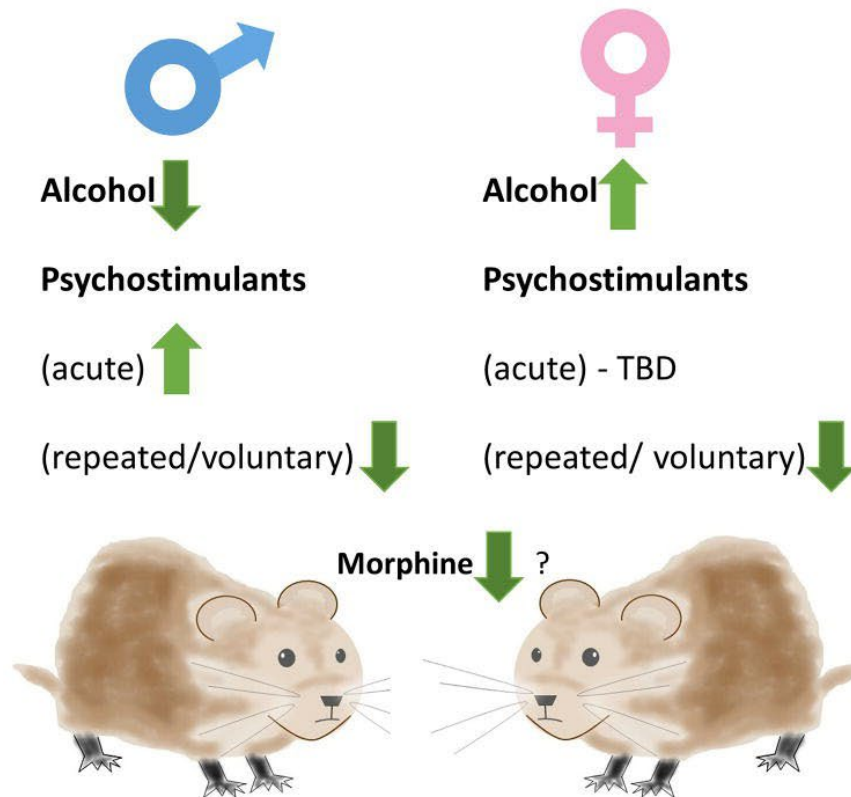
Learning Objective 10.11 Compare the roles of oxytocin and vasopressin in pair bonding.

Pair Bonds Females

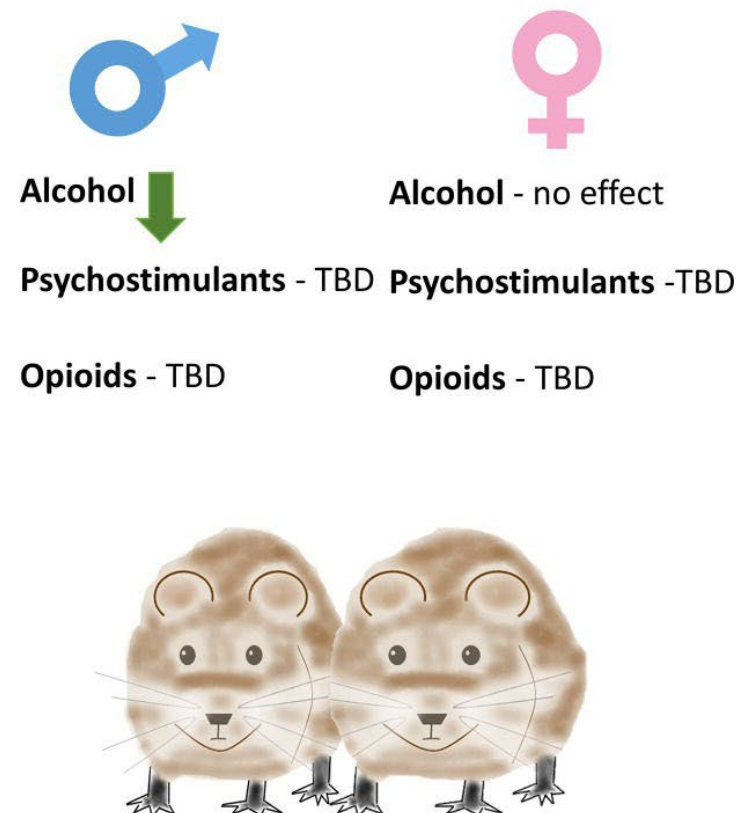
- Mating stimulates release of oxytocin
- Peripheral injection of oxytocin or cerebral ventricles facilitates pair bonding in female prairie voles

The Prairie Vole Model of Pair-Bonding and Its Sensitivity to Addictive Substances

Pair bond formation



Pair bond maintenance



Formation of Pair Bonds

Learning Objective 10.11 Compare the roles of oxytocin and vasopressin in pair bonding.

What is the role of neuropeptides in human pair bonding?

Formation of Pair Bonds

Learning Objective 10.11 Compare the roles of oxytocin and vasopressin in pair bonding.

Humans

- There is an increase in oxytocin after intercourse
 - Feelings of calmness and well-being
- People who received an intranasal spray of oxytocin before looking at photos or faces were more likely to remember these faces later
- Oxytocin administration increases trust, empathy, and altruism but often only for group members or under specific conditions

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Sexual Orientation

- **Roles of Activational and Organizational Effects of Hormones in Sexual Orientation**
- **Role of Steroid Hormones**
- **Sexual Orientation and the Brain**
- **Role of Prenatal Environment in Sexual Orientation**
- **Heredity and Sexual Orientation**

Roles of Activational and Organizational Effects of Hormones in Sexual Orientation

Learning Objective 10.12 Compare the roles of activational and organizational hormone effects in sexual orientation.

How does the exposure of hormones influence sexual orientation?

Roles of Activational and Organizational Effects of Hormones in Sexual Orientation

Learning Objective 10.12 Compare the roles of activational and organizational hormone effects in sexual orientation.

Hormonal Exposure

- Sexual orientation is not strongly related to variations in sex hormone levels in most adults (activational effects)
- Stronger evidence for organizational effects of hormones in sexual orientation

Role of Steroid Hormones

Learning Objective 10.13 Describe examples to support a role of prenatal hormone exposure in sexual orientation.

Prenatal Hormone Exposure

- Shown to affect human social behavior and sexual orientation
 - **Congenital adrenal hyperplasia (CAH)**
 - Higher levels of exposure to androgens
 - Females have higher rate of gay/lesbian orientation
 - Androgen insensitivity syndrome
 - Recent evidence is uncertain
 - Prenatal progesterone exposure

Li, D. et al. Occupational exposure to bisphenol-A (BPA) and the risk of self-reported male sexual dysfunction. Hum. Reprod. 25, 519–527 (2010)

The study included 164 men who were exposed to high levels of BPA in the workplace and 386 men without exposure.

Workers exposed to BPA had a significantly higher likelihood of reduced sexual desire (odds ratio [OR] 3.9, 95% CI 1.8–8.6), erectile dysfunction (OR 4.5, 95% CI 2.1–9.8), ejaculatory dysfunction (OR 7.1, 95% CI 2.9–17.6), and reduced satisfaction with their sex lives (OR 3.9, 95% CI 2.3–6.6; all measured using the International Index of Erectile Function and the Brief Male Sexual Function Inventory).

The researchers also identified a dose–response relationship between BPA exposure and risk of sexual dysfunction. Significant impairment of sexual function was apparent among men whose duration of exposure to BPA was as short as 1 year or less.

Sexual Orientation and the Brain

Learning Objective 10.14 Compare sexually dimorphic features of a male and female brain, and explain what can be learned about the brain basis of sexual orientation from these dimorphisms.

Which structures in the brain show sexual dimorphism?

Sexual Orientation and the Brain

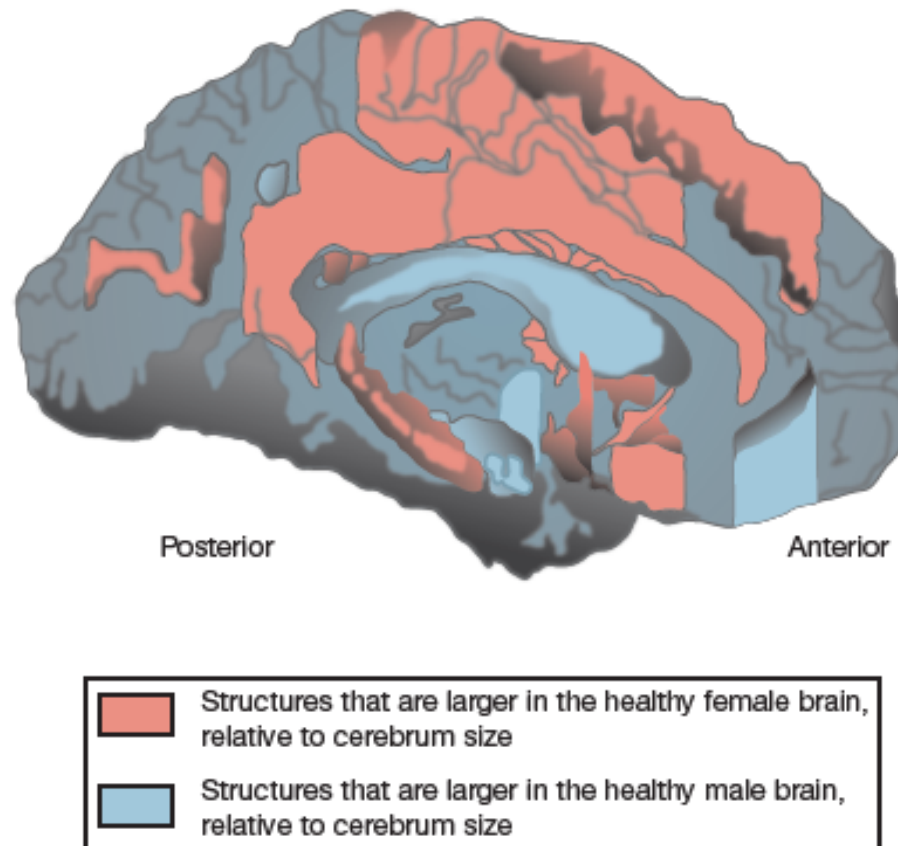
Learning Objective 10.14 Compare sexually dimorphic features of a male and female brain, and explain what can be learned about the brain basis of sexual orientation from these dimorphisms.

Sexual Dimorphism

- Differences in the corpus callosum, hippocampus, amygdala, hypothalamus, prefrontal cortex, and throughout several neurotransmitter and neuroendocrine systems
 - Most believe is result of androgen exposure
- Brains of gay and heterosexual men and women show differences in dimorphic structures
 - Suprachiasmatic nucleus (SCN)
 - Sexually dimorphic nucleus of the hypothalamus
 - Anterior commissure

Figure 10.15

Dimorphic Regions of the Brain



This figure illustrates some of the sexually dimorphic structures in the human brain.

Source: (Goldstein, J. M. et al. Normal sexual dimorphism of the adult human brain assessed by in vivo magnetic resonance imaging. *Cereb. Cortex* 11, 490–497 (2001).)

Role of Prenatal Environment in Sexual Orientation

Learning Objective 10.15 Summarize the relationship between prenatal environmental factors and sexual orientation.

Environmental Factors

- Fraternal birth order effect: gay men tend to have more older brothers
 - No similar effect found in women
 - Mother's immune system may become sensitized to Y-linked proteins present in fetuses

Heredity and Sexual Orientation

Learning Objective 10.16 Summarize what is currently known of the role of heredity in sexual orientation.

Genetic Factors

- Twin studies suggest heredity plays a role in sexual orientation for both men and women
 - Concordance rate for nonheterosexual sexual orientation among identical twin brothers was 52 percent and 22 percent for fraternal twins
 - Concordance for nonheterosexual sexual orientation among female identical twin sisters was 48 percent and 16 percent for fraternal twins

Parental Behavior

- **Maternal Behavior of Rodents**
- **Hormonal Control of Maternal Behavior**
- **Neural Control of Maternal Behavior**
- **Neural Control of Paternal Behavior**

Maternal Behavior of Rodents

Learning Objective 10.17 Describe examples of rodent maternal behavior.

Maternal Behaviors

- Nest building, delivery of pups, cleaning pups, keeping them warm, nursing them, and retrieving them if they are moved out of the nest
- Nest-building behavior is facilitated by progesterone, the principal hormone of pregnancy
- Mothers also induce pups' urination and defecation

Figure 10.16



Beside the nest is a length of rope the mouse used to construct the nest.

Source: Carlson, Neil

Hormonal Control of Maternal Behavior

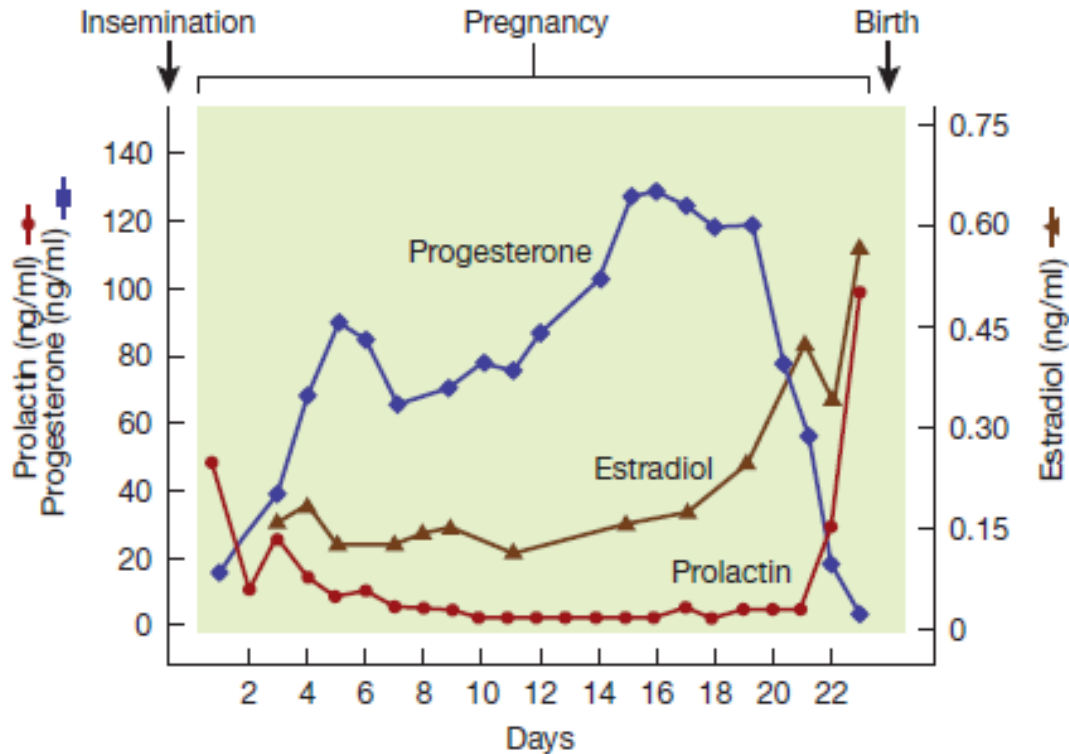
Learning Objective 10.18 Explain the role of hormones in examples of maternal behavior.

Hormonal Influences

- Progesterone, estradiol, and **prolactin** (such as in pregnancy) facilitate maternal behavior
- Oxytocin also facilitates onset of maternal behavior

Figure 10.17

Hormones in Pregnant Rats



The graph shows blood levels of progesterone, estradiol, and prolactin in pregnant rats. Source: (Data from Rosenblatt, J. S., Siegel, H. I., & Mayer, A. D. (1979). Progress in the study of maternal behavior in the rat: Hormonal, nonhormonal, sensory, and developmental aspects. In *Advances in the Study of Behavior* (Vol. 10, pp. 225–311). Academic Press.)

Neural Control of Maternal Behavior

Learning Objective 10.19 Identify brain regions and neural pathways involved in maternal behavior.

Neural Influences

- The medial preoptic area plays a role in maternal behavior
- Injections of progesterone, estradiol, and prolactin that duplicate the sequence that occurs during pregnancy facilitate rat maternal behavior
- fMRI studies show when human mothers look at pictures of their infants, brain regions involved in reinforcement show increased activity

Neural Control of Paternal Behavior

Learning Objective 10.20 Identify brain regions and neural pathways involved in paternal behavior.

Neural Influences

- Paternal behavior is rare in mammal species, but research shows sexual dimorphism of the MPA is less pronounced in voles of monogamous species
 - Lesions of the MPA abolish paternal behavior in male rats
 - Evidence suggests that both prolactin and oxytocin facilitate a human father's role in infant care.

Learning Objectives (1 of 4)

- 10.1** Describe the processes of gamete production and fertilization.
- 10.2** Explain how gonads, internal sex organs, and external genitalia develop.
- 10.3** Compare the organizational effects of hormones on primary sex characteristics and the activational effects on secondary sex characteristics.
- 10.4** Summarize the roles of hormones in phases of the menstrual cycle.
- 10.5** Compare the roles of hormones in sexual behavior of male and female rodents.

Learning Objectives (2 of 4)

- 10.6** Contrast behavioral masculinization and defeminization in rodents.
- 10.7** Compare the activational effects of hormones on sexual behavior of human males and females.
- 10.8** Describe the roles of pheromones in reproductive behavior of humans and other animals.
- 10.9** Identify the roles of spinal and brain mechanisms in male sexual behavior.
- 10.10** Identify the role of brain mechanisms in female sexual behavior.

Learning Objectives (3 of 4)

- 10.11** Compare the roles of oxytocin and vasopressin in pair bonding.
- 10.12** Compare the roles of activational and organizational hormone effects in sexual orientation.
- 10.13** Describe examples to support a role of prenatal hormone exposure in sexual orientation.
- 10.14** Compare sexually dimorphic features of a male and female brain, and explain what can be learned about the brain basis of sexual orientation from these dimorphisms.

Learning Objectives (4 of 4)

- 10.15** Summarize the relationship between prenatal environmental factors and sexual orientation.
- 10.16** Summarize what is currently known of the role of heredity in sexual orientation.
- 10.17** Describe examples of rodent maternal behavior.
- 10.18** Explain the role of hormones in examples of maternal behavior.
- 10.19** Identify brain regions and neural pathways involved in maternal behavior.
- 10.20** Identify brain regions and neural pathways involved in paternal behavior.